GPU Sharing Mechanisms for Multi-Tenant Inference Serving

Group A: Search Execution and Evidence Report

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Abstract

This intermediate deliverable records execution of the submitted SLR protocol and prepares traceable evidence for the final CEP report. The 11 October IEEE/Springer checkpoint contains 689 formal query occurrences and 21 accepted studies. A 14 October extension supplied 23 ACM/ScienceDirect export files with 3,727 parsed occurrences. A supplied 17 October screening register resolves 585 formerly open occurrence decisions and audits assisted exclusions. Most Step 3–5 decisions have not yet been reconciled into the canonical CSV/workbook or a final four-database PRISMA flow; remaining retrieval and inclusion decisions are pending. Batches 1–3 assess 31 supplied queue reports: 22 accepted and nine excluded. Batch 3 adds nine accepted studies and excludes one training-only report. The current accepted set is 43. Source-anchored reviews and receipts are integrated separately from the historical flow. Run-count discrepancies and remaining extraction gaps remain explicit.

Status: evidence checkpoint, not final SLR. The publication window is 1 January 2020–11 October 2026 inclusive. One AI reviewer conducted screening and appraisal at the user's direction; the listed students are the project authors, not claimed independent reviewers. Red text identifies statements, counts or conclusions that must be revisited when the remaining search reconciliation, screening, QA and extraction work is completed.

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1 Purpose and requirements

The engineering problem is to share a physical GPU among inference tenants while balancing useful work and density against predictable latency and isolation. The assigned alternatives are time-slicing, MPS, MIG and API remoting/virtualization, including hybrids. The unit of technical comparison is local sharing on one GPU. A study containing cluster experiments qualifies only when local inference-sharing evidence is identifiable; cluster-wide GPU counts are never substituted for local occupancy.

The submitted `report/SLR_protocol.tex` and `report/SLR_protocol.pdf` are the protocol baseline identified by the user. The protocol contents are preserved. Query deliverables reside in the root `report/` folder; Group A evidence remains under `groups/group-a/evidence/`. The final assignment still requires 15–20 unique peer-reviewed primary studies, at least 15, a 12-point IEEE-cited LaTeX report, and approximately 20–30 pages of main content. Meeting a provisional paper count does not complete searching or the final deliverable. No additional intermediate-page requirement was supplied.

The five questions ask which mechanisms are evaluated (RQ1), how outcomes are measured and compared (RQ2), which operating conditions change results (RQ3), what trade-offs and gaps remain (RQ4), and what conditional choice is supported (RQ5). All five map to extraction and the claim register, rather than to an unconditional ranking of mechanisms.

2 Protocol adoption and deviations

The Group A execution copy adopts the submitted protocol. Amendment GA-A001 extends the cutoff from 27 September to 11 October 2026. GA-A002 records the user's instruction that Codex conduct screening and quality appraisal as a single AI reviewer. There is no independent second reviewer, inter-rater agreement statistic or adjudicated human consensus. GA-A003 permits supplementary full-text work while unavailable database coverage remains open. GA-A004 records the actual public-interface execution and metadata capture in place of signed-in publisher exports. None changes the inclusion criteria to meet a target count.

Eligibility requires an English peer-reviewed original journal or full conference report, a local sharing/partition/process/API mechanism, explicit concurrent inference or justified mixed inference evidence, and sufficient configuration/outcome detail. Training-only, cluster-only, surveys and vendor claims are excluded from the primary set. Unretrieved reports are pending access outcomes, not assessed full-text rejections. Related conference, journal and preprint reports are one underlying study until an independent contribution is established.

3 Review design and operational definitions

This report is the execution record for the adopted Group A protocol, not a second protocol. The decision unit is a unique evaluation of concurrent inference sharing on a physical GPU. An imported search hit is an *occurrence*; matching occurrences are linked to a canonical bibliographic record; a publication version is a *report*; and related reports can describe one *study*. These units are kept separate in every table and flow count. In particular, a server-level GPU saving or an eight-GPU placement outcome cannot be presented as a one-GPU utilization measurement merely because the same paper also evaluates local sharing.

RQ	Analytic question	Evidence needed
RQ1	Which temporal, MPS, MIG, API/runtime and hybrid controls were evaluated?	Mechanism, control layer, local tenant arrangement and comparator.
RQ2	Which outcomes and experimental methods make results comparable?	Metric definition, units, percentile/SLO, denominator, measurement window and baseline.
RQ3	Which workloads, hardware and configurations change the outcome?	GPU/SKU, profile/quota, model and batch, arrivals, co-tenant mix and resource pressure.
RQ4	What engineering costs, isolation properties and contradictory observations are supported?	Within-paper trade-offs, limitations, resource/fault/security experiments and counterexamples.
RQ5	Under which operating requirements is a choice supported?	Conditional synthesis with direct/indirect evidence and explicit unresolved conditions.

The main window is 1 January 2020 through 11 October 2026 inclusive. The publisher's broad 2020–2026 year filter is a retrieval aid; exact dates near the cutoff require checking at the report level. An older seminal exception requires a documented reason in the eligibility record. A result published after the cutoff is not made eligible by appearing in a later export. The exact-date decisions for potentially relevant 2026 reports in the new exports remain to be completed.

3.1 Eligibility decisions and source type

An included primary study must be an English peer-reviewed original journal or full conference report with identifiable concurrent inference evidence, a relevant local GPU-sharing control, and enough configuration and outcome detail to evaluate the research questions. Mixed inference/training studies can contribute only when the inference outcome is separable. Surveys and vendor descriptions may explain terminology but cannot supply an included primary experiment. Reports that cannot be retrieved remain access cases.

At full text, the adopted protocol records one principal reason, with secondary notes where helpful. E1 denotes no inference-serving evidence; E2 no relevant single-node multi-tenant sharing; E3 a mechanism outside scope; E4 training-only; E5 cluster orchestration without separable local sharing; E6 no peer-reviewed original primary report; E7 insufficient technical or outcome detail; E8 a duplicate or less complete report of the same study; E9 a date/language violation without exception; and E10 another written reason. A code is assigned after reading the relevant full report, not inferred from the publisher hit count. The final distribution of full-text reasons will be recalculated after the new reports are assessed.

4 Database-specific search implementation

The four NED-listed platforms cover engineering journals, computing proceedings and related systems literature, but their interfaces differ in fields, syntax and export behavior. The submitted protocol's common concept block combines GPU terminology with inference/model-serving

terminology; the mechanism block covers the assigned families. The recorded executed strings, rather than a retrospectively improved expression, define what was searched. The old five runs and the five later successful runs are separate dated executions. Blocked pilots are retained as attempts with no countable retrieved population.

Platform and family	Executed expression structure	Field/filter and consequence
IEEE MPS, MIG, temporal, API	`(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM)` AND one family block.	All metadata; 2020–2026. The public implementation was not the planned field-qualified Command Search.
Springer master	`(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")`	General search; English, Research or Conference Paper, 2020–2026. Full-text/reference indexing broadens irrelevant matches.
ACM master	`(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")`	All Fields, E-Publication Date 2020–2026; logged Full-Text Collection. Export page overlap/coverage is unresolved.
ScienceDirect temporal, MPS, MIG, API	`(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning")` AND one family block per run.	The master exceeded the connector limit. The fallback omits DNN/LLM and is not equivalent to the other platforms' common block.

The Springer and ACM cells reproduce their identical entered master expression in full. ACM's displayed version wraps each term in an All Fields expression and adds an E-Publication Date clause; the exact transformed string is retained in `search_runs_received.csv`. The IEEE and ScienceDirect cells summarize four family expressions each; their verbatim run-specific strings, source URLs, hits and export receipts are retained in `search_runs.csv`, `search_runs_received.csv` and `export_intake.csv`. The ACM two-file overlap and the ScienceDirect MPS re-export show why a parsed-row count is not itself evidence of complete page coverage. The temporal URL's Open Access facet is documented separately from the

unfiltered count observed later. The final search-coverage statement depends on the ACM export decision, verified filter provenance and any justified amendment; no missing-hit number is imputed.

A 17 October provenance review of an uploaded zip confirmed that all 23 included export files match the receipt ledger by hash and size, and found no screenshot or page-range record in that zip. The zip's file times appear to reflect a later copy and cannot establish the original export times. The helper's separate folder and the live PC directory have not been checked in this review. The two ACM files contain 829 and 1,000 entries with 442 repeated exact keys; overlap suggests page ranges may have intersected, but neither the original ranges nor any number of omitted results is known. The team must record a dated decision to re-export ACM in traceable chunks or retain this coverage limitation. The original run log is preserved, with its received copy differing only by a final CRLF. No query, log or export has been retrospectively changed to reconcile the site counts.

5 Selection, identity and PRISMA accounting rules

Every imported occurrence retains its source file, run and stable ID. An exact DOI or key can propose a link, while a normalized title with a different DOI triggers version review rather than automatic deletion. A conference paper, journal extension and preprint can be several reports of one study. The most complete peer-reviewed report is preferred for claims, while the related versions remain in the history. Whole-proceedings volumes and books in the ACM export need an explicit non-primary-record treatment; their bibliographic keys do not make them individually evaluated papers.

The 11 October checkpoint demonstrates the arithmetic for its captured subset: 689 formal occurrences minus 10 duplicate occurrences plus 20 selected supplementary records yields 699 records considered; 575 earlier-stage exclusions and 96 pending/not-retrieved cases leave 28 assessed reports; seven report-level exclusions leave 21 unique accepted studies. This is a dated checkpoint, not the final four-database flow. A subsequent review note checks all 515 exact DOI/key links, supplies one missing canonical target, and identifies four chains, while the current CSV has not yet been updated. The Step 5 note revises the first-pass classes to 515 repeats, four additional links, 3,130 title/abstract exclusions and 78 advanced/assessed occurrences. These occurrence classes balance at 3,727, but they are not yet canonical report or study counts. The ledger corrections, report-version links and retrieval outcomes must be applied before constructing the merged PRISMA boxes.

Stage	Reconciliation rule and required ledger evidence
Identification	Sum imported occurrences by database/run and supplementary route; retain source file and original display observations separately.
Before screening	Count adjudicated duplicate occurrences and any other removals from named occurrence-to-canonical links; do not remove a report merely for sharing a study ID.
Title/abstract	Count canonical records screened and exclusions with reasons; keep title-only and abstract-informed decisions distinct.

Stage	Reconciliation rule and required ledger evidence
Retrieval	Map records advanced to reports sought; record each not-retrieved report and access attempts without treating it as an assessed exclusion.
Full text	Count assessed reports and one principal exclusion reason per excluded report; link multiple included reports to unique studies.
Inclusion	Show both included reports and unique studies, with study-to-report identity reconciled.

The final four-database PRISMA flow, its reason totals and included-study count must replace the interim figure after the new occurrence ledger is adjudicated. The flow cannot be completed by choosing a convenient difference between a displayed publisher total and an export count. A publisher interface observation and the set of files actually received remain separate provenance facts.

6 Quality appraisal and extraction instrument

Quality scoring is applied to eligible evidence after full-text assessment; it is not a threshold that creates eligibility. Each criterion receives 0, 0.5 or 1, with an individual rationale and source anchor. High (8–10), moderate (6–7.5) and low (below 6) are reporting/interpretability bands. A paper's score is not a security certification, replication claim or proof that its baseline is suitable for all uses.

Item	Question scored from the report
Q1	Is the engineering question and tenant setting explicit?
Q2	Is the mechanism and its configuration sufficiently described?
Q3	Are hardware, software and resource environments identifiable?
Q4	Are workloads, arrival patterns and co-tenant conditions described?
Q5	Is the comparator appropriate and its resource budget clear?
Q6	Are metrics, units, percentiles and denominators defined?
Q7	Are repetitions, warm-up, measurement windows and other controls adequate?
Q8	Does the cited figure/table/prose support the stated result?
Q9	Are validity limits and adverse cases discussed?
Q10	Can the setup be reproduced or at least critically interpreted?

The extraction form links a preferred report and study ID to source type/version, GPU generation and partition or quota, driver/runtime, model and tenant mix, batching/arrival process, comparator, metric and statistic, a bounded outcome, caveat and exact source location. It distinguishes a missing transcription (`NOT EXTRACTED`) from a checked absence in the report (`not reported`). A graph with unreadable pixels is not reverse-engineered into an exact value. The first 21 rows and the two preliminary new assessments are at different maturity levels; newly eligible studies require formal QA, version linkage and a complete anchored extraction before joining the synthesis.

Isolation field	What the observation can and cannot support
Performance interference	Measured latency/throughput under specified co-tenants; cannot alone prove a security boundary.
Resource and memory separation	Profile/quota or memory behavior and measured contention; a partition label alone does not establish an observed guarantee.
Adversarial security	Explicit hostile-tenant or access-control experiment, if reported; otherwise mark not established.
Fault containment	Explicit failure or fault-propagation experiment, if reported; ordinary workload fairness is not a substitute.

The source-linked cards in Appendix A give an auditable QA and extraction view for the current 21-study checkpoint. They are a compact reproduction of the Group A registers, not additional independently reviewed studies. Cards, scores and claims must be regenerated if eligibility, preferred versions or source-grounded extraction changes.

7 Search execution and provenance

NED's E-Resources page lists the four planned publisher platforms. The listing was checked on 11 October 2026: `https://eaklibrary.neduet.edu.pk/eakl/E-Resources.html`. It confirms platform availability in the list, not entitlement to each requested article. No university credentials or licensed full-text PDFs enter this public repository.

Source/run	Actual implementation	Observed hits
IEEE MPS	All metadata; years 2020–2026	25
IEEE MIG	All metadata; years 2020–2026	23
IEEE temporal	All metadata; years 2020–2026	45
IEEE API	All metadata; years 2020–2026	9
Springer master	General search; English; research articles/conference papers;2020–2026	587
ACM pilot (11 October)	Search blocked by challenge loop	Unknown; no import at that checkpoint
ScienceDirect pilot (11 October)	Search page unavailable	Unknown; no import at that checkpoint

The common block is GPU or graphics-processing-unit AND inference, model serving, deep learning, DNN or LLM. IEEE uses one family block per run. Springer uses all family blocks in one OR group. Exact strings, filters, source URLs, dates and export limitations are in `search_runs.csv`. The IEEE field implementation differs from the planned field-qualified Command Search and is disclosed, not described as an equivalent validated command. Research type/language and precise 2026 cutoff checks are manual where facets are unavailable. No accepted study is dated after 2025.

All 30 Springer result pages (20 hits/page,7 on the final page) and all IEEE pages were captured, yielding 587+25+23+45+9=689 occurrences. These are rendered-result metadata imports,

not publisher-generated RIS/CSV exports. Springer export redirects to sign-in; IEEE results export requires personal sign-in and public citation export timed out. The imports retain IDs, database/run, page, title, DOI/URL and displayed date/item metadata. Clock time was not retained and is marked unrecorded.

Springer's broad index includes reference/full-text matches and unrelated acronym expansions such as welding MIG. Search hits therefore do not imply eligible GPU studies. Seed checks recovered ParvaGPU and KRISP in IEEE; papers in other publisher proceedings required supplementary discovery at this checkpoint. Supplementary records are selected named-paper lookups checked after adoption, including submitted seeds and related systems. They are not an exhaustive search-engine result export or completed backward/forward citation search. Their follow-up query strings are reproducible suggestions, not invented historical exact queries.

8 14 October ACM and ScienceDirect extension

The operator supplied 23 database export files and a search log from the named PC export directory. All 23 export SHA-256 hashes were checked against `export_intake.csv` on 16 October and matched. A separate 17 October check of an uploaded repository/export-folder zip likewise found 23/23 matching hashes and byte sizes; this does not verify the live PC directory or a private backup, which remains unconfirmed. The 24th export-folder item is the log, not a database export. The original 6,302-byte log and 6,304-byte `search_log_received.txt` have identical text lines; the received copy adds one final CRLF. The source-file receipt and per-file parser counts are in `export_intake.csv`; the exact received log and structured run transcription are `search_log_received.txt` and `search_runs_received.csv`. The log records ACM's Full-Text Collection and all-fields query, four ScienceDirect family searches, the recorded 2020–2026 year filters, URLs and search times; export times are not recorded. The ACM log states that the run used NED library access, without recording the access mechanism; the operator reported NED VPN use for the ScienceDirect runs. Neither statement establishes entitlement to individual full texts. The protocol's exact 11 October cutoff still applies to individual reports.

Run	Displayed hits	Log export claim	Parsed entries
ACM master	1,600	1,603	1,829
ScienceDirect temporal	1,053	1,053	1,053
ScienceDirect MPS	517	563	563
ScienceDirect MIG	241	241	241
ScienceDirect API	41	41	41
Total of run values	3,452	3,501	3,727

The parsed total consists of 1,829 BibTeX and 1,898 RIS entry *occurrences*, before adjudicated deduplication. The ACM files contain 442 exact keys in both files and 1,387 distinct keys altogether. Of those distinct keys, 472 label whole-proceedings volumes rather than individual papers; their treatment must be recorded in screening. The site showed 1,600 in the saved run register and 1,603 according to the operator; the export contains 1,829 rows. The page-range overlap and completeness of the export are unverified. The numbers represent site observations, parsed rows and keys, respectively, and are not interchangeable. ACM exports used the site's Export button in one session with different page ranges, according to the operator.

ScienceDirect MPS contains 563 parsed entries, including 63 from the separately re-exported `SSD_MPS_06.ris`; 561 distinct DOIs occur in those files. The source of the earlier displayed count of 517 is unrecorded. Reopening the result page on 16 October showed 563 hits, a later observation that supports but does not establish the 14 October page state. The ScienceDirect temporal files contain 1,053 rows and 1,051 distinct DOIs; their `_010`/`_011` filename order was reversed by a naming error, without changing their retained contents. Its logged URL carries an `openaccess` facet absent from the written filter description. On 16 October, the operator observed 298 hits with that facet and 1,053 without it. The 1,053 exported entries are consistent with the unfiltered set, but the filter state at export time cannot be established from this later check. No original log or export is altered to force these counts to agree.

ScienceDirect rejected the intended master string under its reported eight-connector limit, so the four family queries omitted the `DNN OR LLM` terms in the inference block. The received log preserves the exact queries. Family splitting follows protocol Section 5, but dropping terms from the shared inference block is a deviation whose effect on recall is unmeasured. The team has not decided whether to rerun or formally enter an amendment in the protocol register. Any additional content-type or language filter is not recorded. The 2026 year facet alone does not enforce the 11 October cutoff. The five 14 October completed runs are recorded separately from the 11 October blocked pilots. Each log names the first export file for its run; `export_intake.csv` maps all 23 files.

The reproducible import assigns `GA-N00001`–`GA-N03727` in `new_export_occurrences.csv`. Automated DOI/title matching flags nine occurrences against the old 689 formal records, 537 against earlier new occurrences, and twelve candidate links to already accepted study metadata; these classes can overlap and require report-version adjudication. The historical first-pass triage linked 427 exact DOI repeats and 88 exact ACM BibTeX-key repeats, records 2,615 title/abstract exclusions (six individually documented and 2,609 rule-assisted), retained ten priority retrieval leads, separately recorded two assessed eligible full reports, and left 585 topical or ambiguous occurrences for individual title/abstract review. These disjoint first-pass occurrence classes sum to 3,727. Later individual decisions and sensitivity checks are summarized below. Batch 1 now replaces the ten priority retrieval rows with accepted full-text decisions; the remaining note-level outcomes are not yet fully reconciled. The two earlier assessed reports have source-anchored preliminary extraction in `new_fulltext_assessments.csv`, but still need formal QA and version linking before integration into the 21-study synthesis and final flow. A retrieval lead is not an included study. The 21-study tables below preserve the prior checkpoint. Sections 11–13 record the 22 later additions; the current accepted total is 43.

The supplied Step 5 register subsequently gives individual decisions for those 585 occurrences: 55 advance to retrieval, 526 are excluded, and four are linked to existing records or studies. Its sensitivity review checked 43 sharing-related titles among 2,609 rule-assisted exclusions, reinstated eleven for retrieval, and found no error in a seeded sample of 120 from the remainder. The 585 rows plus eleven reinstatements are transcribed as supplemental review input in `received_step5_decisions.csv`; truncated titles in that file are identifiers, not a substitute for the full occurrence ledger. The note therefore reports 78 extension occurrences advanced or already assessed (55+11+12), 3,130 title/abstract exclusions (526+2,604), four further links and 515 exact-key/DOI repeats; these occurrence classes sum to 3,727. The sample does not prove that all remaining assisted exclusions are correct. The supplemental decisions and corrections must be reconciled into the authoritative occurrence, duplicate and screening ledgers before these later counts replace the first-pass workbook and PRISMA view. The two earlier assessed

reports remain outside the accepted count; the later batch 1 update in Section 11 adds ten other studies.

Later batch updates. Ten batch-1 reports were accepted; eleven batch-2 reports were assessed (three accepted and eight excluded). Batch 3 assessed ten more, nine accepted and one excluded. The current accepted set is 43. See Sections 11–13. The queue has 45 unassessed entries and two older preliminary reports still await integration.

9 17 October review-note updates and remaining integration

The supplied Step 1–9 documents extend the earlier export checkpoint. They are review notes with a detailed title/abstract register, not a regenerated authoritative workbook or a complete four-database flow. This distinction is material because the supplied Step 7 flow was written before Step 5: it still calls 585 records open, while Step 5 gives decisions for all 585. The earlier flow is retained as a dated first-pass snapshot; the later decisions take precedence for the current work queue.

Step	Newly documented outcome	Still needed for final flow
1–2	23/23 export hashes and sizes match the uploaded zip; five 14 October runs map to 23 files. ACM page overlap and the ScienceDirect fallback are described.	Confirm private backup/current PC state; decide ACM coverage and query amendment; historical facet state may remain unknown.
3	1,089 parsed 2026 occurrences. Of 21 candidates that could advance, 14 had registry dates inside the cutoff; batch 1 resolves GA-N00946 from its printed event dates, leaving six unresolved.	Confirm consequential dates on publisher pages, especially the six unresolved records.
4	515 first-pass repeat removals were reviewed; 442 repeated ACM keys are contained within the 515. The note supplies the missing canonical target for GA-N01056 and four chain endpoints.	Enter corrected links and version decisions in the canonical ledger; check one same-title different-DOI report pair at full text.
5	All 585 formerly open occurrences have a note-level decision: 55 retrieve, 526 exclude, four link. Eleven rule-assisted exclusions were reinstated after title audit; a 120-title random sample found no further error.	Transfer individual decisions and audit outcomes to source ledgers; sample does not establish zero missed studies.

Step	Newly documented outcome	Still needed for final flow
6–7	The 11 October flow balances at 21 studies. Of 78 extension occurrences advanced or assessed in the Step 5 notes, 33 reports are now assessed: 22 accepted across three batches, nine excluded, and two older eligible reports still pending integration; the supplied 76-entry queue has 45 unassessed.	Reconcile occurrences to unique reports, complete retrieval/eligibility and construct the combined PRISMA diagram.
8–9	The current 43 accepted studies have Q1–Q10 scores and anchored extraction; 22 additions are documented in the three batch registers.	Score new studies only after full-text/version sign-off; verify and transfer any new extraction fields.

The six still-unresolved 2026 date cases are `GA-N00902`, `GA-N00959`, `GA-N00961`, `GA-N00969`, `GA-N01862` and `GA-N01949`. Crossref dates in the supplied note are secondary registry evidence and do not replace a publisher-side check where eligibility turns on an exact date. ScienceDirect's RIS date field appears to mix month or issue stamps; 129 entries with dates after the cutoff were not automatically excluded on that field, and the note reports none among candidates advancing. The six dates, current retrieval outcomes and related-report choices can change the final inclusion and flow.

The second group's QA document demonstrates an extensive presentation format, including a joint score matrix and item-by-item interpretation. Its subject is sandboxed serverless runtimes, with 48 different publications and a seven-item scale. Its study decisions and scores have not been transferred into this GPU-sharing review. The expanded Section 13 below is generated solely from Group A's ten-item `current_quality.csv`; it reports the current 43 studies and their limits without suggesting an independent second appraisal.

Resolved date in batch 1. GA-N00946 now passes the cutoff using its printed SOSP 2026 dates (29 September–2 October), so it is no longer one of the six date cases above.

10 Historical 11 October screening checkpoint and reconciliation

Every formal occurrence receives a record ID. Exact DOI/URL and normalized-title checks link duplicate occurrences without deleting provenance. Bibliographic report IDs identify candidate reports; study IDs identify accepted underlying studies. The REEF journal extension is selected for GA-S001; its less complete conference report is a linked E8 companion and adds no study.

Checkpoint quantity	Count
Formal query occurrences	689
Duplicate occurrences	10
Unique formal records	679
Selected supplementary records	20

Checkpoint quantity	Count
Excluded before full text	575
Full texts assessed	28
Full-text exclusions	7
Accepted unique studies	21
Pending / not retrieved	96

The reconciled captured subset is: 689-10+20=699 records; 699 equals pre-full-text exclusions plus assessed reports plus pending records. Assessed reports equal included reports plus full-text exclusions. The later ACM/ScienceDirect occurrence population is outside these 11 October arithmetic identities. This 11 October checkpoint is intentionally not a completed PRISMA 2020 flow and must be replaced by a reconciled four-database flow. Reports sought include pending retrieval attempts, but complete formal-database retrieval and final study-version resolution remain open.

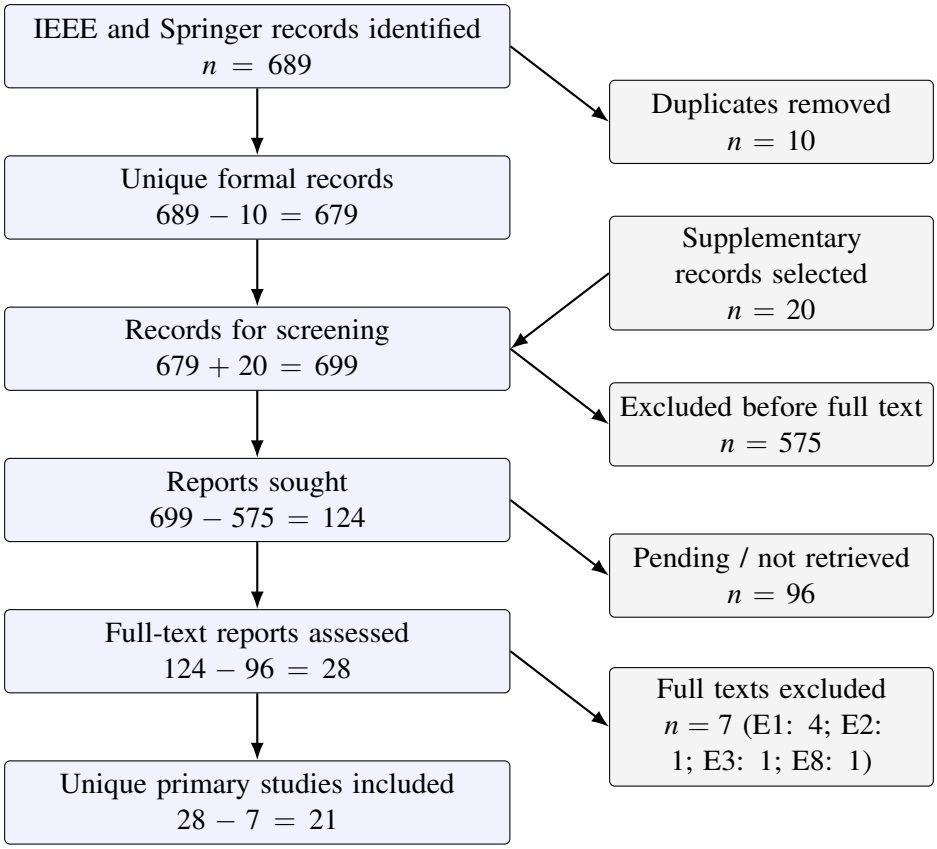


Figure 1: Interim PRISMA 2020 stage mapping for the reconciled 11 October captured subset. The later 3,727 ACM/ScienceDirect occurrences are outside this flow. E1–E8 reason definitions and report-level decisions are in `rejected.csv`. This is not the completed four-database flow.

Title decisions are first-pass relevance judgments, explicitly labelled as title stage. Abstract decisions cite publisher scope; full-text decisions identify a section or experiment. Training and inference are not interchangeable: PipeSwitch and SIRIUS are retained for explicit inference outcomes, whereas TGS is rejected for training-only evidence. The Fermilab paper was retained after its multi-model slice experiment was inspected, rather than merely because it mentioned MIG. DIPPM was rejected because profile prediction alone does not test concurrent tenant sharing.

Full-text rejection examples include DIPPM (no concurrent tenant comparison), the accelerator-pool vRAN simulation and the low-latency FEC offloading paper (no inference evidence), DRMaestro's Rodinia workload evaluation (no serving inference), CalcGraph (assigned GPU partitioning left to future work), and the mediated-device passthrough benchmark (HPC applications without inference). The complete reasons and stages are in `rejected.csv`. Pending files have no fabricated technical results or QA scores.

11 Batch 1: full-text decisions and batch-1 accepted checkpoint

Ten user-supplied full-report conversions were received and assessed on 17 October 2026 by AI-CODEX. All ten satisfy the adopted eligibility rules and add distinct studies, GA-S022–GA-S031. At this batch checkpoint, the accepted evidence set contained **31 unique studies (21+10)**. This is an accepted-set update, not a claim that four-database screening or the final PRISMA flow is complete. Pantheon and Jormungandr retain their earlier eligible-pending-integration status and are outside the 31.

The supplied 76-record deduplicated download queue recorded ten at the batch-1 checkpoint received and assessed reports and 66 then awaiting assessment. The extension full-text register contains twelve assessed reports: the ten accepted here and the two earlier preliminary assessments. Received files are not re-requested. Occurrences, reports, and unique studies remain separate counting units.

11.1 Eligibility, publication window and report identity

The first-page DOI/title matches were checked against the occurrence ledger, the prior 21-study set, and the other reports in this batch. No duplicate study was identified. MIGER, BOER and MMK describe distinct designs and evaluations despite related mechanisms and citations. EnerTune cites an earlier workshop precursor; it is a version lead, not a second included study. All four 2026 reports have printed publication or conference dates before 11 October: MMK (July), MIGServe and AdMix (28 September–1 October), and EnerTune (29 September–2 October). Exact online dates remain distinct from event/issue dates. Deft retains export publication year 2024 and printed WoC 2023 event/reference dates; this discrepancy does not change eligibility.

Deft and AdMix are full original workshop proceedings reports with implemented evaluations, not poster abstracts. Their limited experiment breadth is reflected in appraisal and transfer caveats. Mixed inference/training studies qualify only for their explicit inference-specific local sharing evidence. Multi-GPU aggregate improvements remain separately labelled.

Study	Report	Eligible evidence stratum	QA
GA-S022	miger [1]	MIG partitions with per-partition MPS online/offline co-location, local partition planner and fragmentation control	8
GA-S023	deft [2]	Deft: adaptive CUDA synchronization frequency controlling kernel queues and priority preemption; API interposition prototype	7
GA-S024	fastgshare [3]	FaST-GShare: MPS spatial shares plus CUDA interception/token-based temporal quotas; CUDA IPC model sharing	7.5
GA-S025	split [4]	SPLIT: genetic offline model-block splitting with token-based greedy temporal preemption	8.5
GA-S026	migserve [5]	MIGServe: buddy-aware MIG placement, proactive pair migration and temperature-guided eviction	8
GA-S027	mmk [6]	MMK: MIG, elastic MPS oversubscription and CUDA-intercepted kernel-priority scheduling	8
GA-S028	boer [7]	BOER: MIG model grouping with Bayesian-optimized MPS configurations inside partitions	8.5
GA-S029	admix [8]	AdMix: adaptive MPS active-thread caps, lightweight LLM estimator and best-effort pause/full-access controls	8.5
GA-S030	sgdrc [9]	SGDRC: CUDA/TVM runtime with software VRAM-channel coloring, bimodal tensors and tidal SM masks	8.5
GA-S031	enertune [10]	EnerTune: MIG allocation with joint frequency/batch selection, colocation-power model and runtime tuner; separate MPS portability experiments	9

11.2 Bounded extraction and contradictory evidence

GA-S022 miger. Section 3.1/Fig. 2 directly compares local MPS 57/29/14 percent and MIG 4g/2g/1g allocations. Section 5.2/Fig. 8 reports testbed mean JCT 46.6% below MuxFlow and 36% below MISO; these are job/cluster outcomes, not inference p99 gains. Fig. 9 notes slower small jobs under MIGER. Includes mixed training/inference and multi-GPU optimization; retain only explicitly local sharing and inference-QoS evidence for assigned comparison. Abstract and introduction swap comparator order for the two JCT percentages; use section 5.2 mapping. No chart coordinates reconstructed or universal p99 claim. *Source:* Printed pp. 505-506, 510-511; sections 3.1-3.3, 5.1-5.4; Tables 1-3; Figs. 2, 4, 8-11; N00001.md lines 119-229, 478-662.

GA-S023 deft. Sections 3.2 and 5 show optimal synchronization depends on batch and image size; the fixed strategy can impose much larger preempting-job JCT inflation after a workload change. Section 5.2 reports Deft switches to synchronizing every kernel for batch 30; earlier batch-1 case uses every 500 kernels. Table 2 cells are interleaved, so no exact violation-rate row is imported. Preliminary six-page workshop evaluation with two principal models and one A40. Exhaustive pair profiling does not scale; no cluster/generative-model implementation claimed (section 6). Keep export year 2024 and printed conference/reference year 2023 separately; both inside window. *Source:* Printed pp. 15-17; sections 3.2, 4.2, 5.1-5.3 and 6; Table 2 and Fig. 4 discussed in prose; N00005.md lines 250-468.

GA-S024 fastgshare. Local section 5.3 states eight RNNT pods at 12% SM shares reach about 40 req/s with tail latency below 500 ms, versus one racing pod about 12 req/s and above 1250 ms. The local ResNet/RNNT comparison in Fig. 9 reports removing their observed interference with 24% shares. No unlabelled tail value is relabelled p99. Model-specific/proportional share effects, IPC memory trust and batching limit transfer. Prose fractional-gain notation is ambiguous (3.15x with raw 296.8/71.37), so no such multiplier imported. Supplied plot coordinates/percentile identity and repetitions not reconstructed. *Source:* Printed pp. 641-643; sections 4, 5.1-5.4; Figs. 8-10; N00015.md lines 523-735.

GA-S025 split. Section 5.5 reports violation rate below 10% beyond alpha=4 versus 26% under RT-A in the discussed comparison. For short requests, jitter reductions versus Clockwork/PREMA/RT-A are 55.3/46.8/68.9% at low load and 56.0/50.3/69.3% at high load. Jitter is block-execution standard deviation, not p99 latency. Section 6 reports split overhead and static-graph limits. One edge GPU and synthetic Poisson load; not a server-class GPU generalization. Temporal predictability sacrifices spatial parallelism; long-request stability can worsen. Table 3 split overhead and RNN limits qualify short-request improvements. *Source:* Printed pp. 611-613; sections 5.1-5.5, 6; Tables 1-3; Figs. 6-7; N00040.md lines 451-629.

GA-S026 migserve. Section 5.2 Azure: sustainable rate at 90% SLO attainment is 9.84 RPS versus LA-Dyn 4.13 and Static 1.85. Arena approximately 10.99 versus 2.37 and 2.48 RPS. These are four-GPU server outcomes; local mechanism is MIG layout control, not a single-GPU throughput multiplier. Four-GPU evaluation and constructed layout-agnostic comparator; 90% attainment is weaker than 99% SLO targets used elsewhere. MIG hardware isolation is described, without adversarial/fault experiment. Source pixels and unreported repeats/versions are not reconstructed. *Source:* Printed pp. 850-853; sections 4, 5.1-5.4; Fig. 9 and Figs. 11-13; N00831.md lines 519-780.

GA-S027 mmk. Section 4.3/Fig. 9 reports mean JCT 28% lower, makespan 32% lower and normalized progress-rate STP 35% higher than MIGER. These are mixed-job/testbed metrics, not per-request p99 latency. Section 4.5 reports 1.5% MIG and 3.2% MPS reconfiguration lifecycle overhead. Mixed inference/training and two-GPU optimization; local priority scheduling is the transferable mechanism. Many figure labels contain NUL/glyph corruption, so numerical prose only. STP/JCT does not establish online tail-latency or fault containment. *Source:* Article pp. 84:15-84:19; sections 4.1-4.5; Table 3; Figs. 9-12; N00839.md lines 789-1040.

GA-S028 boer. Local section 2.4/Fig. 3 observes ParvaGPU can underperform MIG-only serving for larger BERT/ResNet50 while helping lightweight MobileNet. Section 5.2/Fig. 8 gives

eight-GPU QoS-preserved throughput mean 34.61% above ParvaGPU; 56.8/62.49/34.63/44.83% above MIG-serving/gpulet/USHER/PARIS. Keep local qualitative counterexample separate from fleet improvements. Bayesian exploration cost and instance interference qualify hybrid gains. Baseline reductions (MIG-serving fast/USHER IR only) and multi-GPU request fan-out constrain comparison. Same senior author as MIGER, but different inference-only system and experiments; not a same-study report extension identified in supplied text. *Source:* Printed pp. 521, 526-528; sections 2.4, 4.6, 5.1-5.4; Tables 2-3; Figs. 3, 8, 10; N00844.md lines 200-265, 700-890.

GA-S029 admix. Section 4.2 reports average best-effort throughput 1.58x Static MPS with local LLaMA-7B (TP1) and 3.9x with Qwen-14B (TP2), keeping strata distinct. AdMix still increases mean TTFT/TPOT versus LLM-only; on Qwen its mean latency increases exceed Static MPS. More throughput is not zero interference. Full six-page original workshop report; one/two-GPU workload strata and tuned-on-test-trace static reference limit transfer. Estimates of average latency do not establish p99. No testbed code rerun or security/fault experiment inferred. *Source:* Printed pp. 3-5; sections 3 and 4.1-4.2; Table 1; Figs. 5-6; N00872.md lines 270-453.

GA-S030 sgdrc. Section 9.3: versus Orion, P40 overall throughput up to 1.47x and BE up to 2.36x; A2000 up to 1.32x and 1.79x, respectively. Mean SLO attainment 99.0% across evaluated settings. Section 10 explicitly states shared-context GPU runtime errors are not isolated. Software coloring is not a proven security boundary. MIG integration is future work; not benchmarked on A100/H100. Reverse-engineering/calibration and TVM compatibility matter; do not treat throughput maxima as averages or infer a P40 MPS result. *Source:* Printed pp. 276-277; sections 9.2-9.3, 10; Table 3 and Fig. 18; N00904.md lines 863-965.

GA-S031 enertune. Section 2.1/Fig. 2: co-location choice can increase energy; SDXL plus MobileNetV3 example uses 13% less energy versus no sharing despite 1.14x latency. Section 7.1/Fig. 9: 16-GPU testbed energy reduced by factors 1.4-2.3 and power by 1.3-2.6 versus baselines while meeting SLOs; fleet results are not local per-GPU savings. Joint power/energy objectives select different allocations. Local model-pair evidence must stay distinct from fleet savings and from frequency-only effects. Source chart pixels not available; experiments/code not rerun; missing warm-up and independent-run uncertainty not assumed. *Source:* Printed pp. 66-67, 73-75; sections 2.1-2.2, 6, 7.1-7.2; Figs. 2, 9; N00946.md lines 140-249, 798-995.

11.3 Quality, isolation and sensitivity

The ten batch scores total 81.5/100. At the 31-study batch-1 checkpoint, the total is 255.5/310 (mean 8.24); 25 high, 6 moderate and 0 low. The original Q1–Q10 rubric is unchanged. Each new item score has a rationale and source anchor in `batch1_quality.csv`; `current_quality.csv` combines the historical and batch registers. The evidence cards included 31 studies at this checkpoint; batch 2 later updates the combined cards.

Excluding moderate-QA reports removes Deft and FaST-GShare from this batch as well as the four moderate reports in the earlier set. This weakens adaptive synchronization and serverless quota evidence; it does not establish that all remaining studies have comparable or complete measurement controls. No numerical effect is pooled across studies.

SGDRC explicitly cannot contain co-located GPU runtime faults in its shared context (section 10). MIG-based partition descriptions do not establish adversarial security or fault-injection results. AdMix and EnerTune show why higher throughput or utilization is insufficient: protected-service latency and energy can worsen under particular sharing configurations. These are conditional trade-offs, not universal rankings.

11.4 Evidence and remaining work

The historical 11 October registers and flow remain available unchanged. Extension decisions are authoritative in `new_fulltext_assessments.csv`, supported by `batch1_review_assessments.json`, per-item QA, extraction and SHA-256 receipts. The combined accepted views are `current_accepted.csv`, `current_extraction.csv`, `current_quality.csv` and `current_included_studies.json`. The workbook Included and Quality tabs showed 31 studies at this checkpoint; its Summary tab is explicitly historical and Current shows the later accepted set. The bibliography contained 31 preferred reports at this checkpoint.

Complete the remaining retrieval/eligibility records, reconcile the supplied Step 3–5 decisions and duplicate/version links, integrate or resolve the two earlier extension assessments, and derive a complete four-database PRISMA flow. The 66 entries remaining at this checkpoint are candidate reports, not guaranteed additions. No unavailable chart data, reviewer consensus, or screening outcome is invented.

12 Batch 2: full-text reassessment

Eleven further user-supplied full-report conversions from the deduplicated queue were assessed on 17 October 2026 by the same single AI reviewer. Three add unique studies GA-S032–GA-S034; eight are excluded with principal full-text reasons. Prior rule-assisted title/abstract exclusions were reconsidered when the full texts arrived. The accepted set is now **34 unique studies (21 historical + 10 batch 1 + 3 batch 2)**; this does not complete merged database screening or PRISMA accounting.

The 76-entry download queue has 21 received reports (13 accepted, eight excluded) and 55 unassessed entries. The extension full-text ledger has 23 assessed reports, including two older eligible-pending-integration reports outside the 34. Source hashes, reasons and anchors are retained without committing paper bodies.

12.1 Decisions and evidential bounds

Record	Decision	Primary reason or direct evidence
GA-N00920	Accept GA-S032 [11]	Include: original HPDC full paper evaluates concurrent inference functions with local MIG partitioning and request SLO outcomes; separate server-level from local mechanism claims.
GA-N01085	Accept GA-S033 [12]	Include: original EuroSys full paper explicitly evaluates co-located inference instances with an implemented single-A100 temporal/MPS hybrid.

Record	Decision	Primary reason or direct evidence
GA-N01096	Accept GA-S034 [13]	Include as direct interference/isolation evidence: real single-GPU LLM inference is concurrently run with a distinct synthetic process/kernel under streams and Green Context SM partition; a mixed co-tenant experiment is explicitly bounded, with no claim of two production inference services.
GA-N00002	Exclude E2	Cricket/unikernel CUDA remoting is evaluated with microbenchmarks and proxy applications across two nodes, without a concurrent inference-tenant GPU comparison. The JetsonInference library appears in related work, not the evaluated setting.
GA-N00004	Exclude E1	AWS Lambda/rCUDA case study compares CPU, native GPU and remote GPU for a single GPU-BLAST processing path; AI inference is motivation, without measured inference-serving outcomes for concurrent tenants.
GA-N00055	Exclude E4	nvshare's measured co-located TensorFlow ResNet152v2 applications train models; the authors explicitly select training-only evaluation although inference is mentioned as background.
GA-N00088	Exclude E1	GPUPool evaluates throughput-oriented long jobs and simulated kernel pairs; inference appears as a motivating workload class/reference, without an inference-specific measured serving outcome.
GA-N00102	Exclude E2	Guardian's multi-application spatial-sharing comparison uses epoch-based training/co-run workloads (Table 2, Fig. 6); inference phases are tested standalone for overhead (Fig. 7b). There is no evaluated concurrent inference-serving arrangement under the adopted criterion.
GA-N01195	Exclude E2	ReFINE evaluates synthetic priority kernel pairs in Accel-Sim architectural simulation rather than a measured concurrent inference arrangement on one physical GPU; inference serving is an illustrative motivation.
GA-N01866	Exclude E1	Edge-device rCUDA evaluation runs the Fuzzy Minimals clustering algorithm across remote node/GPU settings; no inference-serving workload or outcome is evaluated.
GA-N01945	Exclude E1	MIG scheduling and reconfiguration evaluates Rodinia/Altis benchmarks and synthetic task queues for makespan; inference latency is that of the DRL scheduler itself, not a model-serving tenant outcome.

GA-S032 fluidfaas. Section 7.1/Fig. 9 reports FluidFaaS SLO hit-rate improvement of 90% for medium and 61% for heavy workload mixes over the compared systems; the abstract reports 25–75% throughput increases across workloads. These are system-level results, not a measured per-GPU utilization effect. Pipeline, placement and MIG allocation contribute jointly; comparisons with a non-pipelined baseline cannot isolate the benefit of MIG partitioning alone. No adversarial security or fault-injection evidence is inferred from MIG descriptions. *Anchor:* N00920.md abstract, sections 3, 6, 7.1; Tables 2, 5–6; Figs. 9 and 12.

GA-S033 bless. The introduction and evaluation report mean latency reductions of 39.1% for two BERT inference instances and 41.2% for four, under the paper's compared GPU-sharing systems. Other reported aggregate gains are not recast as p99 or universal speedups. Quota, model pairing and GPU generation condition results. The conversion has damaged column/table layout, so no unlabeled plotted coordinates or precise tail-percentile gains were recovered. *Anchor:* N01085.md intro, sections 4, 6.1–6; Fig. 1 and BERT co-location result prose.

GA-S034 interference_deeper. H100 Llama decode P90 time between tokens rises from 7.53 to 16.56 ms under a sleep co-tenant in section 4.2 (batch 1, 1,000-token prompt). Table 1's Green Context experiment isolates SM sets (64/68) but reports P90 TBT as high as 18.38 ms under copy-kernel memory pressure; its alone baseline is 14.2 ms on the whole GPU, so partition cost and interference cannot be collapsed into one effect. Synthetic co-tenants and short decode traces quantify resource contention; the paper proposes a future scheduler but does not implement an end-to-end serving controller. Green Context SM partition does not partition L2 or external memory bandwidth. Table 1 conversion is interleaved; only prose and clearly attributable values are used. *Anchor:* N01096.md sections 3–5, especially 4.1–4.4; Fig. 2 and Table 1.

12.2 Appraisal and remaining reconciliation

The three new studies score 24/30; the combined 34 score 279.5/340, with 28 high, 6 moderate, and 0 low. The same Q1–Q10 rubric applies only to included studies. Source-specific Q-item rationales and limitations are in `batch2_quality.csv` and `batch2_extraction.csv`.

The SoCC paper prints a November 2025 event date although the export lists 2026; both lie within the cutoff. It measures synthetic kernel co-tenants alongside LLM inference, not two production inference services. Guardian shows co-located training and standalone inference only; it therefore fails the adopted concurrent inference criterion despite useful memory-safety context.

Resolve 55 remaining queue entries, the two older eligible-pending-integration reports, report-version and occurrence links, and the four-database PRISMA flow. Excluded batch-2 reports are assessed full texts, not additions to the historical Flow A seven full-text exclusions.

13 Batch 3: full-text review

Ten further user-supplied full-report conversions were assessed by one AI reviewer on 17 October 2026. Nine add unique studies GA-S035–GA-S043; one is excluded. The accepted set is now **43 unique studies (21 historical + 10 batch 1 + 3 batch 2 + 9 batch 3)**. This is not a reconciled

merged-database PRISMA inclusion count. The 76-entry queue has 31 assessed reports and 45 unassessed entries. Two older eligible-pending-integration reports remain outside the 43.

Record	Decision	Reason or bounded finding
GA-N00027	Accept GA-S035 [14]	Include: original journal evaluation has concurrent inference on local edge GPU, with heterogeneous results interpreted separately.
GA-N00046	Accept GA-S036 [15]	Include: original full conference paper measures concurrent inference and critical-task latency on edge GPUs.
GA-N00048	Accept GA-S037 [16]	Include: original full conference evaluation of concurrent tenants on a shared GPU with inference latency/goodput outcomes.
GA-N00049	Accept GA-S038 [17]	Include: section 5.4 directly evaluates two inference jobs sharing one GPU; mixed training experiments are qualified separately.
GA-N00059	Accept GA-S039 [18]	Include: original full proceedings evaluation of simultaneous transformer inference tenants sharing GPU resources.
GA-N00063	Accept GA-S040 [19]	Include with scope qualification: original paper implements local MIG/MPS inference sharing and evaluates concurrent inference, while synthesis separates system-level outcomes.
GA-N00068	Accept GA-S041 [20]	Include with scope qualification: measured local shared-GPU inference pairs occur on one server; aggregate energy is contextual only.
GA-N00070	Accept GA-S042 [21]	Include: original full paper tests concurrent inference and local MPS spatial resource allocations.
GA-N00073	Accept GA-S043 [22]	Include with scope qualification: original full paper evaluates local GPU sharing of inference functions embedded in a serverless workflow.
GA-N00018	Exclude E4	IADeep evaluates co-located iterative deep-learning training jobs and training throughput. Inference is not an evaluated concurrent tenant workload under the adopted inference criterion.

GA-S035 edge_sky. Abstract and multi-tenancy evaluation report inference throughput gains under concurrent execution; do not attribute heterogeneous placement gains solely to GPU sharing. Edge GPU, CPU and TPU contributions are coupled; throughput is not latency predictability or security isolation. *Anchor:* N00027.md abstract; multi-tenancy and performance evaluation sections, figures on CME and DMP.

GA-S036 miriam. Abstract reports up to 92% throughput improvement while critical tasks incur less than 10% latency overhead in its evaluated scenarios. Edge elastic-kernel scheduling is

transferable temporal/spatial evidence, not a direct MIG/MPS comparison. *Anchor:* N00046.md abstract; sections on kernel coordination and evaluation.

GA-S037 elasticroom. The evaluated engine balances request goodput, throughput and latency under co-located inference; comparison is engine-level. Compiler and priority scheduling effects cannot be isolated as pure partition effects; plotted numbers were not transcribed. *Anchor:* N00048.md abstract; section 6.2 evaluation and Fig. 1.

GA-S038 switchflow. Section 5.4/Fig. 10 reports interleaved execution gains among inference jobs of up to 30% versus session-based time slicing in tested pairings. The 19.05x p95 inference latency example in Fig. 6 uses a training co-tenant and is not evidence for two inference tenants. *Anchor:* N00049.md sections 5.1, 5.2.1 and 5.4; Figs. 6 and 10.

GA-S039 itif. Abstract reports up to 2.40x performance versus baselines for evaluated shared-backbone transformer inference. Shared backbone changes model computation; speedup cannot be assigned to GPU scheduling alone or generalized to unrelated models. *Anchor:* N00059.md abstract; design and experiment sections including four-tenant setup.

GA-S040 protean. The evaluation reports SLO compliance and cost effects of orchestration under trace-driven inference; no per-GPU improvement is extracted from system-level totals. Multiple GPUs and global scheduling contribute jointly; local mechanism claims cannot be isolated without matched per-device comparisons. *Anchor:* N00063.md abstract, sections on MIG/MPS design and evaluation of 22 workloads.

GA-S041 wattwiser. Section 4/Fig. 3 shows model tail latency relative to SLO while requests consolidate onto shared GPUs; power savings are whole-server effects. The four-GPU power outcome cannot estimate per-GPU partition efficiency; MI50 setup is not a MIG evaluation. *Anchor:* N00068.md sections 4.1–4.2 and Fig. 3.

GA-S042 deinfer. Section 5 evaluates prediction accuracy and allocation for concurrent inference with MPS; its Fig. 2 shows that MPS spatial limits do not eliminate shared-resource latency interference. Two GPUs are present in testbed; claims here concern within-device workloads, not a two-device pooled speedup. *Anchor:* N00070.md sections 2.2 and 5.1–5.2; Fig. 2, Table 1 and testbed table.

GA-S043 esg. Abstract reports 61–80% SLO hit-rate improvement and 47–187% cost savings over prior systems for evaluated workflows. Figures are workflow/system results influenced by batching and scheduler search, not isolated effects of local GPU sharing. *Anchor:* N00073.md abstract, sections 3 and 5; evaluation setup and end-to-end comparison.

The batch-3 reports extend the assessed evidence, not the historical Flow A boxes. Reconcile the remaining 45 queue entries, two older eligible reports, duplicate/version links, database searches and the combined PRISMA flow.

14 Historical 21-study accepted checkpoint

The following studies are accepted for this intermediate evidence set. The study list, preferred reports and QA totals may change after the new search population is screened and version links are settled. KRISP is a scope-boundary inclusion under protocol 7.2 because it evaluates local spatial resource control and inference workers; its emulated AMD feature and non-NVIDIA baseline are explicitly isolated in synthesis. No vendor document or preprint-only unpublished work counts toward this table.

Study	Report/year	Mechanism / scope	QA
GA-S001	REEF 2025 [23]	Reset-based temporal preemption + dynamic kernel padding; NVIDIA binary instrumentation	8.5
GA-S002	GPULET 2022 [24]	MPS + temporal	8.5
GA-S003	PIPESWITCH 2020 [25]	Temporal context switching	9.0
GA-S004	CLOCKWORK 2020 [26]	Temporal exclusive scheduling	8.5
GA-S005	TORPOR 2025 [27]	API remoting + temporal swapping	8.5
GA-S006	STREAMBOX 2024 [28]	API/runtime sandbox virtualization	8.0
GA-S007	KRYPTON 2025 [29]	Kernel API virtualization + hybrid	9.0
GA-S008	SIRIUS 2025 [30]	Temporal + MPS memory handover	8.5
GA-S009	TALLY 2025 [31]	CUDA API interception + priority	9.0
GA-S010	PARVA 2024 [32]	MIG + MPS	8.0
GA-S011	ORION 2024 [33]	CUDA API interception + temporal	9.0
GA-S012	GSLICE 2020 [34]	MPS adaptive provisioning	9.0
GA-S013	KRISP 2023 [35]	Spatial partition/API control (AMD)	8.0
GA-S014	XSCHED 2025 [36]	Temporal API interposition	9.0
GA-S015	PAELLA 2023 [37]	Temporal software scheduling + MPS comparison	8.5
GA-S016	GNNTRITON 2024 [38]	MIG + concurrent model serving	8.5
GA-S017	VASTPIPE 2025 [39]	MPS adaptive quotas; separate AMD CU-mask stratum	6.5
GA-S018	ARKGPU 2023 [40]	CUDA API interception + NVML monitoring and admission throttling	7.0
GA-S019	DYNAP 2025 [41]	MIG dynamic placement/reconfiguration with reconfiguration-last policy	7.0
GA-S020	IGNITER 2023 [42]	MPS active-thread quotas + interference-aware batch/provisioning and shadow process	8.5
GA-S021	INFERFAIR 2024 [43]	Temporal SLO-aware scheduling + feedback goodput balancing; MPS comparator	7.5

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copies are linked to verified proceedings metadata and do not become separate preprint studies. ParvaGPU's author manuscript is provisional for publisher-version comparison; its reported runtime-version anomaly remains flagged. The supplied REEF journal conversion establishes a more complete implementation/evaluation; it is now the preferred report for GA-S001. The conference report is retained as a linked companion. Accessible copies of Orion, GSLICE, KRISP, Tally and Paella avoid unnecessary institutional download requests.

14.1 Source-anchored finding register

Study	Bounded finding	Source location
GA-S001 [23]	GV100 PyTorch DISB-C: Orion p 99 normalized latency 1.17x vs Reef below 1.05x; normalized total throughput 0.97x vs 1.14x(Fig 29). TensorRT comparison trades throughput: mean 1.21x Triton vs 1.11x Reef(Fig 28). All ratios retain model-solo normalization; do not mix engines/AMD strata.	Article pp 1:23-1:24,1:31-1:34; sections 6.2,6.6,7; Figs 28-29; Markdown lines 1111-1180,1600-1820
GA-S002 [24]	Local gpulets support interference-aware provisioning; cluster throughput is a different outcome.	PDF p 10 Table 2/3; section 5; local profiling section 3
GA-S003 [25]	Fig 7: inference preemption latency reduced by up to 38.2ms.	PDF pp 11-13; sections 6-6.2; Fig 7
GA-S004 [26]	Fig 5 measures goodput satisfying SLO, rather than raw request throughput.	PDF p 10; section 6.1; Fig 5
GA-S005 [27]	BERT Table 4: native 42ms, remote 43ms, PCIe swap 144ms, NVLink swap 45ms.	PDF pp 10-11; Table 4; sections 7.2-7.3
GA-S006 [28]	Table 2 evaluates runtime sandbox workflows on shared GPUs.	PDF p 10 onward; section 9; Table 2
GA-S007 [29]	Inference-specific sharing comparisons; section 2.2 tests fault propagation.	PDF p 10 Tables 1-2; section 5; section 2.2 Fig 3
GA-S008 [30]	Inference p 99/SLO outcomes accompany training progress; separate TTFT/TBT LLM case.	Section 6.1 Fig 10; section 6.6
GA-S009 [31]	At 10% idle time: normalized throughput Tally 1.19 vs TGS1.65.	PDF pp 9-11; sections 5-5.3; Fig 6
GA-S010 [32]	Fig 7 separates internal/external fragmentation; Fig 5 counts GPUs.	PDF p 8 onward; sectionIV; TableIV; Figs 5/7
GA-S011 [33]	Interference-aware scheduling compares inference latency and BE throughput.	PDF pp 8-9; sections 5.3-6.1; Tables 1/3
GA-S012 [34]	GUE measures efficiency; shadow instance loading takes about 12s in Fig 15.	PDF pp 8-12; section 4; Table 3; Fig 15
GA-S013 [35]	Kernel-wise right-sizing evaluates throughput and p 95 inference latency.	PDF pp 9-10; sectionsV-VI; TableIII

Study	Bounded finding	Source location
GA-S014 [36]	Fig 15: relative p 99 1.53/1.51/1.07 respectively versus solo.	PDF pp 14-15; section 8 Case 3 Fig 15; section 9
GA-S015 [37]	p 99 latency and throughput under bursty model mixes; Fig 12 includes MPS.	PDF pp 9-12; section 7 Table 3/Figs 11-12; section 8
GA-S016 [38]	Fig 10: thrashing after two models/20GB or five/40GB.	HTML Multi-model Scaling/Fig 10; Triton implementation; limitations
GA-S017 [39]	A100 VGG16+GoogLeNet: throughput 2.7x no MPS and 1.5x default MPS; not a universal average or p 99 result. NLP/CV mean 2.43x no MPS, 1.45x default MPS. MI100 mean 1.89x no mask, 1.24x default mask.	Published pp 457-459; sections 6.7-6.9; Figs 14-16; uploaded Markdown lines 1150-1290
GA-S018 [40]	Section 5.2 reports all 30 tested pairs satisfy the 2x-solo p 99 QoS target with ArkGPU. Low-contention pairs can lose about 5% QPS from throttling. Four-GPU app-mix goodput gains 94.98% vs k 8s-native and 38.65% vs MPS are separate system-level results.	Published pp 313-319; sections 4.3,5.1-5.4; Table 3; Figs 14-21; Markdown lines 920-1360
GA-S019 [41]	Single-A30 batch experiments: smaller partitions increase concurrency but can increase mean JCT; not online p 99 evidence. Fig 8 broader job set reports 39.03% lower makespan vs StaCon,12.14% vs MISO and 14.7% higher normalized STP vs StaCon. Prediction-error sensitivity:15% perturbation adds 8.09% makespan.	PDF pp 11-16; sections 4.1-4.3.2; Table 2; Figs 8-13; Markdown lines 700-865
GA-S020 [42]	Local four-model latency prediction error 1.53%-5.02%(Fig 13). Fig 14 deployment: iGniter 6 GPUs/0 SLO violations vs gpu-lets+8/3,FFD+5/10,GSLICE+6/3;25% lower hourly instance cost vs gpu-lets+, not 25% higher local utilization. Fig 17 shows an initial p 99 violation before shadow-process recovery.	Published pp 820-824; sections 4.2,5.1-5.4; Figs 13-18; Markdown lines 938-1245
GA-S021 [43]	Local Table 3 load 3:1:2: unfairness 1.05 vs Clockwork 1.50; SLO goodput 524.29 vs 457.24req/s. Other load ratios give smaller differences. Local unfairness is max/min model slowdown(Eq 2), not p 99 or security. Four-GPU MAF outcome kept separate.	Published pp 17-18; sections 5.1-5.4; Table 3; Eq 2; Markdown lines 718-899

15 Review of six supplied full texts

Five reports add unique studies: VastPipe, ArkGPU, Dyna-P, iGniter and InferFair. REEF's journal report explicitly extends OSDI2022 with a complete NVIDIA implementation, additional kernel handling and newer comparisons; the journal is preferred within GA-S001. Six previously pending reports are now assessed; the earlier conference report becomes a full-text E8 companion. There are 21 accepted studies,28 assessed reports including 7 full-text exclusions, and 96 pending access cases. Search occurrence and discovery counts are unchanged.

The source files are Markdown conversions. Two-column interleaving, missing plot pixels and damaged mathematical symbols prevent exact extraction of some equations, CDFs, error bars and configurations. Only readable labelled tables, captions and explicitly stated numeric results enter the register. Missing values are not interpolated. The receipt ledger records filenames and SHA-256 hashes; licensed source contents are not published.

Local GPU strata determine inclusion. VastPipe's NPU-cluster outcomes, ArkGPU's four-GPU goodput, Dyna-P's multi-GPU aggregates and cluster simulation, iGniter's fleet cost, and InferFair's four-GPU trace are distinguished from local results. ArkGPU qualifies through the explicit mixed inference/training exception; transfer is limited to priority inference protection. Dyna-P's job completion and makespan do not establish online p 99. InferFair's unfairness is a relative slowdown ratio and does not establish adversarial isolation.

Journal REEF's same-engine GV100 PyTorch comparison reports DISB-C p 99 below 1.05 times solo versus 1.17 for Orion, alongside separately normalized total throughput. Its TensorRT comparison also reports lower mean normalized throughput than Triton; preserving this counterexample prevents a universal superiority claim. iGniter's initial p 99 violation followed by shadow-process correction likewise qualifies any steady-state SLO statement. Source locations and QA rationales are recorded per report.

At the stricter QA threshold 8, VastPipe, ArkGPU, Dyna-P and InferFair are omitted; their mechanism-specific conclusions weaken. At the protocol low-quality threshold 6, none is omitted. This is a reporting-quality sensitivity, not a pooled causal result. The set now exceeds the protocol's 15–20-study target range; retain all eligible studies at this checkpoint and document any final narrowing transparently rather than removing evidence to fit a count.

16 Quality appraisal and extraction controls

16.1 Section 13: item-level quality assessment

The adopted ten-item rubric scores reporting and interpretability from 0 (absent), 0.5 (partial) or 1 (adequate), for a maximum of 10. Eligibility was assessed separately. The preferred peer-reviewed report of each accepted study is scored once; the earlier Reef conference version remains in the assessment history but does not add a study. The same single AI reviewer prepared all scores. No independent agreement statistic or experimental replication is claimed.

Study	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	/10	Band
GA-S001	1	1	0.5	1	1	1	0.5	0.5	1	1	8.5	high
GA-S002	1	1	1	1	0.5	1	0.5	1	0.5	1	8.5	high
GA-S003	1	1	0.5	1	1	1	1	1	0.5	1	9	high
GA-S004	1	1	0.5	1	0.5	1	0.5	1	1	1	8.5	high
GA-S005	1	1	0.5	1	1	1	0.5	1	0.5	1	8.5	high
GA-S006	1	1	0.5	1	0.5	1	0.5	1	0.5	1	8	high

Study	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	/10	Band
GA-S007	1	1	1	1	0.5	1	0.5	1	1	1	9	high
GA-S008	1	1	0.5	1	0.5	1	1	1	0.5	1	8.5	high
GA-S009	1	1	0.5	1	1	1	0.5	1	1	1	9	high
GA-S010	1	1	0.5	1	0.5	1	0.5	1	0.5	1	8	high
GA-S011	1	1	0.5	1	0.5	1	1	1	1	1	9	high
GA-S012	1	1	0.5	1	1	1	1	1	0.5	1	9	high
GA-S013	1	1	1	1	0.5	1	0.5	1	0.5	0.5	8	high
GA-S014	1	1	0.5	1	1	1	0.5	1	1	1	9	high
GA-S015	1	1	0.5	1	0.5	1	0.5	1	1	1	8.5	high
GA-S016	1	1	0.5	1	0.5	1	0.5	1	1	1	8.5	high
GA-S017	1	1	0.5	1	0.5	1	0	0.5	0.5	0.5	6.5	moderate
GA-S018	1	1	1	1	0.5	1	0	0.5	0.5	0.5	7	moderate
GA-S019	1	1	0.5	1	0.5	1	0.5	0.5	0.5	0.5	7	moderate
GA-S020	1	1	0.5	1	0.5	1	1	0.5	1	1	8.5	high
GA-S021	1	1	0.5	1	0.5	1	0.5	1	0.5	0.5	7.5	moderate
GA-S022	1	1	0.5	1	0.5	1	0.5	1	1	0.5	8	high
GA-S023	1	1	0.5	1	0.5	1	0	0.5	1	0.5	7	moderate
GA-S024	1	1	0.5	1	0.5	0.5	0.5	1	0.5	1	7.5	moderate
GA-S025	1	1	0.5	1	0.5	1	0.5	1	1	1	8.5	high
GA-S026	1	1	0.5	1	0.5	1	0.5	1	0.5	1	8	high
GA-S027	1	1	1	1	0.5	1	0.5	0.5	0.5	1	8	high
GA-S028	1	1	0.5	1	0.5	1	0.5	1	1	1	8.5	high
GA-S029	1	1	0.5	1	0.5	1	0.5	1	1	1	8.5	high
GA-S030	1	1	0.5	1	0.5	1	0.5	1	1	1	8.5	high
GA-S031	1	1	1	1	0.5	1	0.5	1	1	1	9	high
GA-S032	1	1	0.5	1	0.5	1	0.5	1	1	0.5	8	high
GA-S033	1	1	1	1	0.5	1	0.5	1	0.5	0.5	8	high
GA-S034	1	1	1	0.5	0.5	1	0.5	1	1	0.5	8	high
GA-S035	1	1	0.5	0.5	0.5	1	0.5	1	0.5	0.5	7	moderate
GA-S036	1	1	0.5	1	0.5	1	0.5	1	1	0.5	8	high
GA-S037	1	1	0.5	1	1	1	0.5	1	0.5	0.5	8	high
GA-S038	1	1	1	1	1	1	0.5	1	1	1	9.5	high
GA-S039	1	1	0.5	1	0.5	1	0.5	1	1	0.5	8	high
GA-S040	1	1	1	1	0.5	1	0.5	1	1	0.5	8.5	high
GA-S041	1	1	1	1	1	1	0.5	1	1	1	9.5	high
GA-S042	1	1	1	1	0.5	1	0.5	1	1	1	9	high
GA-S043	1	1	0.5	1	0.5	1	0.5	1	1	0.5	8	high

The 43 preferred reports total 355 points (mean 8.26, range 6.5–9.5). There are 36 high (8–10), 7 moderate (6–7.5) and 0 low (below 6) scores. Their individual Q-item rationales, preferred-report IDs and source anchors are recorded in `current_quality.csv`; Appendix A presents study-level evidence cards.

Item	Interpretation	Adequate	Partial	Absent
Q1	Question and tenant setting	43	0	0
Q2	Mechanism and configuration	43	0	0
Q3	Hardware/software profile	12	31	0
Q4	Model, load and co-tenant mix	41	2	0
Q5	Comparator and resource budget	9	34	0
Q6	Metric definition and units	42	1	0
Q7	Measurement controls	5	35	3
Q8	Support in figure/table/text	36	7	0
Q9	Threats and adverse cases	25	18	0
Q10	Reproduction or critical interpretation	27	16	0

Hardware/software completeness, baseline fidelity and measurement controls remain qualified across the current set. The item-count table above is regenerated from all 43 rows. Numeric

prose can support a bounded finding even when the supplied conversion lacks readable plot pixels; no missing curve or confidence interval is reconstructed.

Sensitivity and version control. Removing the 7 moderate/low scored reports leaves 36 high-scored studies. This weakens particular mechanism comparisons and does not prove complete causal controls in the retained set. The preferred Reef journal extension remains one study, with its conference assessment retained as history.

Limits of the appraisal. Some supplied conversions lacked readable chart images; Q8 is based on accessible captions, tables and stated numbers, not reverse-engineered pixels. The experiments and code were not rerun, so Q10 assesses whether a reader can critically interpret the configuration. Performance isolation, resource separation, adversarial security and fault containment need different experiments; a high total cannot establish all four. Pantheon and Jormungandr remain unscored pending report-version and eligibility sign-off; if admitted, their own source-anchored Q1–Q10 rows and all summary counts must be regenerated. The QA matrix and evidence cards now use the combined 43-study accepted set; the historical 21-study tables above are retained as an explicitly dated checkpoint.

The extraction register captures source/version, mechanism, hardware/runtime, tenant workload, comparator, a bounded finding, caveat and precise source location. Fields not yet fully transcribed are labelled NOT EXTRACTED, distinct from a source's checked not-reported value. The supplied Step 9 notes give a per-study narrative for the 21-study checkpoint and preliminary Pantheon/Jormungandr entries; they do not add a study or turn an unavailable figure into a numeric result. Any useful new transcription must be checked against the preferred report and entered into the authoritative extraction register before the report, workbook and claims are regenerated.

Isolation observations remain four different columns: performance interference, resource/memory separation, adversarial security and fault containment. Most current findings establish workload performance rather than comparative security or fault containment. The KRYPTON fault experiment is specifically identified; it does not validate all mechanisms or all GPU generations.

16.2 Section 14: data extraction status after the extension

The extraction and appraisal matrices for the 21 accepted studies were prepared from the previously assessed source texts. No new ACM or ScienceDirect occurrence has been added to these matrices solely on the strength of bibliographic metadata. Two ACM full proceedings reports have now been assessed directly from author-hosted copies: Pantheon (MobiSys 2024) evaluates concurrent inference scheduling on Jetson GPUs and reports deadline misses of 0.39–1.10% in its three application mixes (PDF section 5.1, Fig. 9); Jormungandr (SYSTOR 2024) profiles one-A100 multi-model MIG/MPS/batch configurations but evaluates allocation on an eight-A100 cluster (PDF sections 3.2 and 8, Figs. 3 and 6–7). These are provisionally eligible, with source-anchored fields and transfer limits in `new_fulltext_assessments.csv`; formal QA, report-version linkage and synthesis integration remain open. The supplied Step 5 notes identify 78 extension occurrences advanced or already assessed, including these two; ten original priority retrieval leads were assessed in batch 1, then eleven additional queued reports in batch 2 and ten in batch 3. The supplied queue now has 45 unassessed entries. Title/abstract exclusions have no full-text or extraction claims. The workbook's affected batch rows are

updated in `NewQueue`, `Triage` and `AssessedNew`; the other Step 3–5 decisions still require canonical reconciliation. `Current`, `Included` and `Quality` show the 43-study accepted set. All three batches now have anchored extraction and QA; remaining candidates and parameter gaps prevent completion of the full Section 14 evidence pool.

17 Provisional answers to the research questions

The following RQ answers are a synthesis of the current 21-study checkpoint. Their coverage and confidence must be checked against the final eligible set and completed extraction.

17.1 RQ1: mechanism and control boundaries

The set contains temporal scheduling, MPS resource provisioning, MIG plus MPS, and API/runtime interception or remoting. Hybrids are counted as studies once, not once per mechanism. Software scheduling may control task admission, kernel dispatch, preemption or memory handover; these are different operating boundaries. A process-sharing prototype, a kernel-intercepted container and a hardware slice cannot be assigned the same security meaning. The detailed mapping is in `extraction.csv` and `claims.md`.

17.2 RQ2: outcome definitions and comparability

Keep requests/second, SLO-compliant goodput, allocated GPU count, GPU busy time, SM occupancy, memory utilization and paper-specific efficiency ratios separate. Likewise p 95, p 98, p 99, average latency, time-to-first-token and time-between-tokens are not substitute measurements. A reduction in context-switch latency does not by itself establish p 99 end-to-end latency isolation. Normalize only within an explicitly comparable experiment and retain the original denominator and statistic.

Comparison check	Required condition before numerical comparison
Hardware	Same GPU/vendor/memory/profile or explicit indirect label; separate AMD emulation from NVIDIA mechanisms.
Workload	Same models/input sizes, batching, tenants and co-tenant mix; mixed training comparisons retain inference outcomes.
Load	Same open/closed-loop arrival process, rate, burstiness, warm-up and measurement window.
Comparator	Same resource budget; disclose reimplementations, wrappers and baseline adaptations.
Outcome	Same latency percentile/SLO and throughput denominator; occupancy and device-count reduction remain separate.

17.3 RQ3: operating conditions

Current evidence suggests that model memory demand, launch granularity, arrival burstiness, batching, priority mix and transfer path alter the sharing result. GPU generation constrains which resource boundaries are available. Local slice experiments are informative for memory

pressure, while cluster scaling results require their own placement/network interpretation. Recommendation confidence falls when driver/runtime, quota/profile and experiment windows have not yet been completely transcribed.

17.4 RQ4: trade-offs and counterevidence

Stricter priority protection may trade lower-priority throughput for predictability. Remoting or swapping adds path-dependent transfer cost. Resource partitioning may reduce packing flexibility. Fine-grained control can require compiler changes, API interception, driver hooks or baseline adaptations. These implementation costs must accompany reported gains. The source-anchored counterexample and parameter-specific results reside in the extraction register, avoiding an invented common percentage improvement across studies.

17.5 RQ5: conditional engineering choices

The following are provisional reviewer interpretations, not validated deployment instructions. The candidate/evidence links must be refreshed after final selection and QA sensitivity analysis.

Operating objective	Candidate to investigate	Evidence / unresolved condition
Priority-sensitive inference	Temporal/API scheduling with admission and memory controls	GA-S001, S003, S004, S008, S009, S011, S014; direct per-paper tests, indirect transfer.
Density with controlled workload mix	Tuned MPS provisioning	GA-S002, 012; check memory/compute contention and reconfiguration cost.
Slice boundary with concurrent models	MIG, optionally MPS inside a slice	GA-S010, 016; profile, packing and memory pressure; no universal isolation result.
Serverless remoting and elasticity	API remoting/runtime virtualization	GA-S005, 006,007; runtime compatibility and transport budget.
Strong adversarial/fault guarantee	No universal selection supported	Comparative evidence insufficient; retrieve security candidates and distinguish failure types.

18 Sensitivity, limitations and continuation

Removing QA<6 removes no current study, so that sensitivity test is uninformative. Removing provisional author-manuscript evidence weakens hybrid MIG and the recorded negative throughput comparison. Excluding AMD-only feature evidence limits cross-vendor conclusions. Removing all incompletely transcribed environment profiles would sharply reduce evidence;

final recommendations require completing those fields. No pooled effect or universal winner is asserted.

The main limitations are unresolved ACM/ScienceDirect export coverage and count provenance, remaining Step 3–5 screening notes not yet integrated into the canonical ledgers/workbook, single-AI-reviewer bias, inaccessible full texts, supplementary-selection bias, baseline and hardware heterogeneity, incomplete parameter extraction and unresolved report versions. The accessible set is concentrated in systems proceedings and older DNN workloads; LLM and adversarial isolation evidence requires further checking. A minimum paper count cannot offset these limitations.

Next, enter the supplied Step 3–5 decisions into canonical ledgers, settle export and duplicate/version links, assess the 45 remaining entries in the supplied deduplicated download queue, complete parameter extraction and rerun claims/sensitivity. The user supplied six of the ten earlier requested papers; all six are assessed. The remaining four are unavailable and no further paper download is requested for that earlier batch. Link any new report to its underlying study before adding a citation. Six previously unresolved 2026 date cases remain after the batch 1 printed-date check resolves GA-N00946. Scope changes still require a documented amendment and instructor guidance; this report does not relax eligibility.

19 Evidence package and institutional requests

The evidence directory supplies search runs, imported records, duplicates, all decisions and reasons, accepted/rejected/pending views, retrieval lookups, QA, extraction, claims and checkpoint counts. `query_references.bib` contains the 34 currently accepted preferred primary reports, each cited here. The spreadsheet is a review view; CSV/JSON records remain the authority. The Current tab and current-prefixed registers combine the original 21, batch 1 ten and batch 2 three new studies; the Summary tab preserves the historical Flow A arithmetic. Scripts validate linkage and reconciliation and regenerate derived views.

The unresolved access backlog is in `download_requests.csv`; there are zero active requests. `download_requests.md` records the six received conversions and four unavailable original requests. Twenty-one reports across batches 1 and 2 have since been received and assessed; the updated 76-entry download queue has 55 entries remaining. The original four unavailable requests are still not repeated. All requested publisher platforms appear on NED's list, but article-level access must be checked using university access. PDFs should be supplied privately through the conversation, not committed to the public repository. The supplied export files preserve a new occurrence ledger, while exact syntax, collection, facets and conflicting displayed/export totals remain recorded for reconciliation.

A Study-level evidence cards

These cards reflect the current 43-study accepted set after batch 3; the complete four-database selection flow remains open. The item rationales below reproduce the Group A quality register. Bounded outcomes and caveats reproduce the extraction register. They are not a new independent appraisal or a substitute for the cited full texts.

A.1 GA-S001: Real-time, Work-conserving GPU Scheduling for Concurrent DNN Inference

Preferred report: ACM Transactions on Computer Systems, 2025; report GA-PX007. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Reset-based temporal preemption + dynamic kernel padding; NVIDIA binary instrumentation
Environment	One MI50, 16GB/60CUs; i7-10700, 32GB RAM; Ubuntu20.04, ROCm4.3.0, TVM0.14.0. Separate one GV100, 32GB/80SMs; CUDA11.4, PyTorch1.12.0, TensorRT8.5.1. Driver version NOT EXTRACTED.
Workload	DISB A-E, uniform RT except Poisson E; closed-loop BE; ResNet152, VGG19, YOLOv3, BERT-base, GPT2 (128 tokens, GPT2 prefill). RT rates50%,100% or10% solo-throughput according to workload; Apollo/SVL trace. Separate LLM case batch1,100 decode tokens, MLC-LLM.
Comparator	RT-only; sequential priority; GPU streams. NVIDIA closed-source strata compare priority-enabled Triton/TensorRT and Orion/PyTorch without dynamic padding.
Bounded finding	GV100 PyTorch DISB-C: Orion p99 normalized latency1.17x vs Reef below1.05x; normalized total throughput0.97x vs1.14x(Fig29). TensorRT comparison trades throughput: mean1.21x Triton vs1.11x Reef(Fig28). All ratios retain model-solo normalization; do not mix engines/AMD strata.
Transfer limit	Explicit OSDI2022 extension; select journal as preferred report and retain GA-S001 once. Requires deterministic/stateless kernels; closed-source comparison omits padding. Memory assumed sufficient; bandwidth contention and input-shape profiling limits. Printed publication date differs from some issue-year indexing; recorded2025 follows supplied copy and is in cutoff either way.
Source anchor	Article pp1:23-1:24, 1:31-1:34; sections6.2, 6.6, 7; Figs28-29; Markdown lines1111-1180, 1600-1820

QA	Score	Source-based rationale
Q1	1	RT and BE sharing question explicit

QA	Score	Source-based rationale
Q2	1	Reset/preemption, padding and NVIDIA implementation described
Q3	0.5	GPU/CPU/RAM/OS/frameworks clear; driver missing
Q4	1	DISB mix, RT relative rates, arrivals and model/token configurations explicit
Q5	1	Same-engine closed-source comparisons and priority settings disclosed
Q6	1	Latency and throughput denominators explicit; p99 prose anchored toFig29
Q7	0.5	Solo reference averaged1000 runs; repeats/duration for all serving tests unresolved
Q8	0.5	Captions/numerical comparisons survive; full CDF/plot data absent
Q9	1	Section7 details statelessness, determinism, graphs, bandwidth and memory limits
Q10	1	Implementation and explicit engine constraints permit interpretation; experiments not rerun

A.2 GA-S002: Serving Heterogeneous Machine Learning Models on Multi-GPU Servers with Spatio-Temporal Sharing

Preferred report: USENIX ATC, 2022; report GA-PS002. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	MPS + temporal
Environment	RTX2080Ti 11GB; two/server; Ubuntu18.04; CUDA10.2; driver440.64; PyTorch1.10
Workload	Nine models; Poisson arrivals; per-model SLO
Comparator	Adapted SBP/Nexus; best-fit with/without interference
Bounded finding	Local gpulets support interference-aware provisioning; cluster throughput is a different outcome.
Transfer limit	Server-level gains cannot be treated as one-GPU utilization.
Source anchor	PDF p10 Table2/3; section5; local profiling section3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	1	Environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls

QA	Score	Source-based rationale
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.3 GA-S003: PipeSwitch: Fast Pipelined Context Switching for Deep Learning Applications

Preferred report: USENIX OSDI, 2020; report GA-PS003. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal context switching
Environment	V10016GB or T416GB EC2; PyTorch1.3; CUDA10.1
Workload	Training batch32; inference8; RN152/InceptionV3/BERT; 100 runs
Comparator	Unoptimized/grouped/per-layer transfer
Bounded finding	Fig7: inference preemption latency reduced by up to 38.2ms.
Transfer limit	Average switch/inference latency is not p99; multi-VM throughput separate.
Source anchor	PDF pp11-13; sections6-6.2; Fig7

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.4 GA-S004: Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

Preferred report: USENIX OSDI, 2020; report GA-PS004. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal exclusive scheduling
Environment	V10032GB; DellR740; worker32cores/768GB; two GPUs/host
Workload	Local experiment: one GPU; 15 RN50 copies; 16 clients/model; SLO10-500ms
Comparator	Clockwork variants; INFaaS on different hardware
Bounded finding	Fig5 measures goodput satisfying SLO, rather than raw request throughput.
Transfer limit	Cross-hardware INFaaS comparison is indirect; controller/worker distinction.
Source anchor	PDF p10; section6.1; Fig5

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.5 GA-S005: Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

Preferred report: USENIX ATC, 2025; report GA-PS005. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	API remoting + temporal swapping
Environment	Worker48vCPU/384GB; four V10032GB
Workload	Eight inference models; warm-up; CV80ms/BERT200ms SLO
Comparator	Native; remote GPU; PCIe/NVLink swap
Bounded finding	BERT Table4: native42ms, remote43ms, PCIe swap144ms, NVLink swap45ms.
Transfer limit	p98 tail; busy time is not SM occupancy; node scaling separate.

Extraction field	Recorded observation
Source anchor	PDF pp10-11; Table4; sections7.2-7.3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.6 GA-S006: StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

Preferred report: USENIX ATC, 2024; report GA-PS006. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	API/runtime sandbox virtualization
Environment	Four V10016GB; 28cores/128GB; CUDA10.1; PyTorch1.3; TensorFlow2.12
Workload	Three concurrent inference workflows/GPU
Comparator	MPS; INFless; Astraea; Stream-only
Bounded finding	Table2 evaluates runtime sandbox workflows on shared GPUs.
Transfer limit	Sandbox memory separation is not proof of adversarial security.
Source anchor	PDF p10 onward; section9; Table2

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table

QA	Score	Source-based rationale
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.7 GA-S007: KRYPTON: Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception

Preferred report: USENIX ATC, 2025; report GA-PS007. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	Kernel API virtualization + hybrid
Environment	A10040GB/RTX4090; Ubuntu20.04; kernel5.15.0-91; driver550.90.07; CUDA12.1; PyTorch2.2.1
Workload	Inference/training/rendering in separate containers; solo-throughput targets
Comparator	Temporal; MIG; adapted gpulet
Bounded finding	Inference-specific sharing comparisons; section2.2 tests fault propagation.
Transfer limit	Baseline adaptations and workload mix constrain attribution.
Source anchor	PDF p10 Tables1-2; section5; section2.2 Fig3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	1	Environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.8 GA-S008: SIRIUS: Colocating ML Inference and Training with Fast GPU Memory Handover

Preferred report: USENIX ATC, 2025; report GA-PS008. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal + MPS memory handover
Environment	V10016GB; Ubuntu20.04; CUDA11.6; PyTorch2.1.2; TVM0.14; LLM case A10080GB
Workload	One GPU; six DNNs/up to56 instances + SWIN training; 300s; Poisson MAF; SLO4x solo
Comparator	Reimplemented TaskSwitch; SP50/SP75; UM+MPS
Bounded finding	Inference p99/SLO outcomes accompany training progress; separate TTFT/TBT LLM case.
Transfer limit	TaskSwitch is a reimplementation, not vanilla PipeSwitch.
Source anchor	Section6.1 Fig10; section6.6

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.9 GA-S009: Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads

Preferred report: ACM ASPLOS Volume 1, 2025; report GA-PS009. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	CUDA API interception + priority
Environment	A10040GB; Ubuntu20.04; CUDA12.2; PyTorch2.2; ONNX1.17
Workload	Same-GPU pairs; MAF inference traces; high-priority p99
Comparator	Time slicing; MPS; priority MPS; TGS
Bounded finding	At 10% idle time: normalized throughput Tally1.19 vs TGS1.65.

Extraction field	Recorded observation
Transfer limit	Latency protection sacrifices some BE throughput; author manuscript.
Source anchor	PDF pp9-11; sections5-5.3; Fig6

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.10 GA-S010: ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

Preferred report: IEEE/ACM SC, 2024; report GA-P0589. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	MIG + MPS
Environment	A10080GB; Ubuntu20.04; CUDA12.0.1; reported PyTorch1.14.0 needs checking
Workload	Eleven models; six scenario mixes; same-model MPS inside MIG
Comparator	MIG-serving; gpulet and system variants
Bounded finding	Fig7 separates internal/external fragmentation; Fig5 counts GPUs.
Transfer limit	Author manuscript; unusual runtime version; GPU count is not occupancy.
Source anchor	PDF p8 onward; sectionIV; TableIV; Figs5/7

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear

QA	Score	Source-based rationale
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.11 GA-S011: Orion: Interference-aware Fine-grained GPU Sharing for ML Applications

Preferred report: ACM EuroSys, 2024; report GA-PS011. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	CUDA API interception + temporal
Environment	V10016GB/CUDA10.2; A10040GB/CUDA11.3; PyTorch1.12; Python3.9
Workload	Inference-inference and inference-training; uniform/Poisson; three repeats
Comparator	MPS; streams; temporal; REEF-N reimplementation
Bounded finding	Interference-aware scheduling compares inference latency and BE throughput.
Transfer limit	Prototype clients share a process; REEF-N differs from AMD REEF.
Source anchor	PDF pp8-9; sections5.3-6.1; Tables1/3

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.12 GA-S012: GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

Preferred report: ACM SoCC, 2020; report GA-PS012. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	MPS adaptive provisioning
Environment	One V10016GB; XeonGold6148; kernel4.4.0-150; DPDK19.02; CNTK/PyTorch/TensorRT
Workload	Two-four inference functions; VGG case SLO50ms; traffic replay
Comparator	Standalone functions; MPS provisioning variants
Bounded finding	GUE measures efficiency; shadow instance loading takes about12s in Fig15.
Transfer limit	GUE is not GPU busy percentage; median latency not p99; academic mirror.
Source anchor	PDF pp8-12; section4; Table3; Fig15

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	1	Measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	1	Interpretability; experiments not rerun

A.13 GA-S013: KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers

Preferred report: IEEE HPCA, 2023; report GA-P0594. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	Spatial partition/API control (AMD)
Environment	One MI50/60CUs; dualEPYC7302/512GB; Ubuntu18.04; kernel5.4; ROCm4.5
Workload	One/two/four workers and heterogeneous pairs; batch32; aggregate8/16; eight models
Comparator	AMD unrestricted concurrency labelled MPSDefault

Extraction field	Recorded observation
Bounded finding	Kernel-wise right-sizing evaluates throughput and p95 inference latency.
Transfer limit	Feature emulated; overhead subtracted; baseline is not NVIDIA MPS.
Source anchor	PDF pp9-10; sectionsV-VI; TableIII

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	1	Environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	0.5	Partial: validity limits
Q10	0.5	Partial: interpretability; experiments not rerun

A.14 GA-S014: XSched: Preemptive Scheduling for Diverse XPUs

Preferred report: USENIX OSDI, 2025; report GA-PS014. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal API interposition
Environment	One GV100 inference case; broader XPU experiments separate
Workload	Two BERT-large clients; HP10req/s; saturated BE
Comparator	Triton; Triton+priority; Triton+XSched
Bounded finding	Fig15: relative p99 1.53/1.51/1.07 respectively versus solo.
Transfer limit	Data assumed to fit memory; CUDA graphs and undocumented driver limits.
Source anchor	PDF pp14-15; section8 Case3 Fig15; section9

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	1	Comparator suitability/configuration
Q6	1	Units/latency statistic clear

QA	Score	Source-based rationale
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.15 GA-S015: Paella: Low-latency Model Serving with Software-defined GPU Scheduling

Preferred report: ACM SOSP, 2023; report GA-PS015. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Temporal software scheduling + MPS comparison
Environment	TeslaT4; XeonSilver4114/64GB; Linux4.15.0; CUDA11.7; modified TVM0.10
Workload	Eight models; lognormal sigma1.5/2; stabilized measurements
Comparator	CUDA single/multiple streams; MPS; Triton23.03; ablations
Bounded finding	p99 latency and throughput under bursty model mixes; Fig12 includes MPS.
Transfer limit	Custom Triton TVM wrapper; MPS omitted from Fig11; dynamic batching unsupported.
Source anchor	PDF pp9-12; section7 Table3/Figs11-12; section8

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.16 GA-S016: Optimizing High-Throughput Inference on Graph Neural Networks at Shared Computing Facilities with the NVIDIA Triton Inference Server

Preferred report: Computing and Software for Big Science, 2024; report GA-P0041. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	MIG + concurrent model serving
Environment	A10080GB; 20/40GB slices; two hosts/four GPUs each; AMD Epyc7543; 100Gbps
Workload	Distinct ParticleNet copies within one MIG slice; four requesting workers/model; 120s samples
Comparator	20GB vs40GB slices; scaling settings
Bounded finding	Fig10: thrashing after two models/20GB or five/40GB.
Transfer limit	Local slice evidence only; multi-server scaling separate; isolation properties described, not attack-tested.
Source anchor	HTML Multi-model Scaling/Fig10; Triton implementation; limitations

QA	Score	Source-based rationale
Q1	1	Explicit tenant question
Q2	1	Mechanism specified
Q3	0.5	Partial: environment completeness
Q4	1	Workload specified
Q5	0.5	Partial: comparator suitability/configuration
Q6	1	Units/latency statistic clear
Q7	0.5	Partial: measurement controls
Q8	1	Result figure/table
Q9	1	Validity limits
Q10	1	Interpretability; experiments not rerun

A.17 GA-S017: VastPipe: A High-Throughput Inference System via Adaptive Space-Division Multiplexing for Diverse Accelerators

Preferred report: Journal of Computer Science and Technology, 2025; report GA-P0006. **Current QA:** 6.5/10 (moderate).

Extraction field	Recorded observation
Mechanism	MPS adaptive quotas; separate AMD CU-mask stratum

Extraction field	Recorded observation
Environment	Local NVIDIA A100: TensorRT serving, MPS; AMD MI100: TVM/ROCm with hipExtStreamCreateWithCUMask. GPU memory, OS, driver and runtime versions NOT EXTRACTED for these GPU experiments. NPU Table3 is not the GPU configuration.
Workload	Co-located CV pairs (VGG16/GoogLeNet, VGG19/GoogLeNet, ResNet101/InceptionV3, VGG19/ResNet50); NLP/CV pairs with BERT/GPT2; Poisson-generated request mixes. Exact GPU batch sizes and rates NOT EXTRACTED.
Comparator	No MPS; default MPS without per-service quotas; RL-selected MPS quotas. AMD no/default CU mask separately.
Bounded finding	A100 VGG16+GoogLeNet: throughput 2.7x no MPS and 1.5x default MPS; not a universal average or p99 result. NLP/CV mean 2.43x no MPS, 1.45x default MPS. MI100 mean 1.89x no mask, 1.24x default mask.
Transfer limit	Include only local GPU inference evidence, not the main 1000-NPU cluster results. Offline Q-table assumes relatively stable request mix; GPU-specific quotas, repetitions and variance unresolved. Chart bars are not preserved in Markdown. Preliminary ICCD2021 report is linked as an unassessed version lead.
Source anchor	Published pp457-459; sections6.7-6.9; Figs14-16; uploaded Markdown lines1150-1290

QA	Score	Source-based rationale
Q1	1	Local model co-location question explicit in6.7/6.8
Q2	1	Per-model MPS control and CU-mask API specified
Q3	0.5	GPU models/backends known; GPU OS/driver/version/memory unresolved
Q4	1	Named co-located models and request distribution; exact load incomplete
Q5	0.5	No/default/tuned sharing compares same platform; exact quotas incomplete
Q6	1	Throughput ratios and requests/s labelled; no tail-latency inference
Q7	0	GPU-specific warm-up, run duration, repetitions and variance not recoverable from supplied text
Q8	0.5	Figures/captions and numerical prose survive; chart bars do not
Q9	0.5	Stable-mix/Q-table assumptions stated; GPU threats incompletely discussed
Q10	0.5	GPU results interpretable with caveats; insufficient configuration for replication

A.18 GA-S018: ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs

Preferred report: CCF Transactions on High Performance Computing, 2023; report GA-P0037.
Current QA: 7.0/10 (moderate).

Extraction field	Recorded observation
Mechanism	CUDA API interception + NVML monitoring and admission throttling
Environment	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5. Per-GPU memory/framework versions NOT EXTRACTED.
Workload	Five MLPerf inference LC models (ResNet50, SSD-MobileNet, SSD-ResNet34, DLRM, BERT), each paired locally with six BE training models (30 pairs). Single-stream query size1; p99 QoS threshold twice solo. Separate four-GPU app-mixes are not local occupancy evidence.
Comparator	MPS (Non-ArkGPU); k8s-native for QPS/utilization; GaiaGPU and adjustment-unit ablation. MPS quota settings NOT EXTRACTED.
Bounded finding	Section5.2 reports all30 tested pairs satisfy the 2x-solo p99 QoS target with ArkGPU. Low-contention pairs can lose about5% QPS from throttling. Four-GPU app-mix goodput gains94.98% vs k8s-native and38.65% vs MPS are separate system-level results.
Transfer limit	Mixed inference/training qualifies via explicit inference p99/QPS and workload transfer rationale; it does not establish inference-only performance. Polling/API sleeps can add delay; BE checkpoint/restart affects costs. Huge goodput ratios can have near-zero baseline yield. Repetitions/variance and chart coordinates unresolved.
Source anchor	Published pp313-319; sections4.3, 5.1-5.4; Table3; Figs14-21; Markdown lines920-1360

QA	Score	Source-based rationale
Q1	1	LC protection during BE co-location explicit
Q2	1	CUDA interception, NVML polling, sleep and adjustment described in4.3
Q3	1	Table3 specifies GPU/CPU/OS/driver/CUDA/cuDNN; memory not transcribed
Q4	1	MLPerf single-stream, query size1, 30 pairs and QoS rule explicit
Q5	0.5	MPS/native/GaiaGPU compared; quotas and baseline implementation incomplete
Q6	1	p99, average, QPS and yield-based goodput distinguished
Q7	0	Warm-up/window/repetitions/uncertainty not recoverable in supplied text
Q8	0.5	Tables and numerical prose readable; plot data unavailable

QA	Score	Source-based rationale
Q9	0.5	Throttling overhead and low-contention regressions discussed; limited threats analysis
Q10	0.5	Algorithm and data-on-request statement; workload/configuration gaps restrict replication

A.19 GA-S019: Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs

Preferred report: Cluster Computing, 2025; report GA-P0014. **Current QA:** 7.0/10 (moderate).

Extraction field	Recorded observation
Mechanism	MIG dynamic placement/reconfiguration with reconfiguration-last policy
Environment	Two A30s, 24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon profiles2g.12gb+1g.6gb+1g.6gb. OS/backend versions NOT EXTRACTED.
Workload	50 SLM inference jobs; batch1-64; DistilGPT, Codegen, Flan-T5-Large, Codellama and Phi-1 as labelled in Table2. Local batch sensitivity:10 inference jobs, batches1-8 vs32-64 on one A30. Later160-A100 cluster study is simulation.
Comparator	Full-GPU sequential baseline; fixed StaCon MIG; pre-profiled MISO adapted to A30; ORACLE with future job/size/time knowledge.
Bounded finding	Single-A30 batch experiments: smaller partitions increase concurrency but can increase mean JCT; not online p99 evidence. Fig8 broader job set reports39.03% lower makespan vs StaCon, 12.14% vs MISO and14.7% higher normalized STP vs StaCon. Prediction-error sensitivity:15% perturbation adds8.09% makespan.
Transfer limit	Retain one-GPU inference strata separately from two-GPU aggregate and cluster simulation. JCT/makespan/STP cannot replace request-tail latency or goodput. Reconfiguration can stop/restart active jobs; neither architectural MIG claims nor OOM avoidance prove hostile isolation. Model labels/parameter counts and corrupted overhead fraction require caution.
Source anchor	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865

QA	Score	Source-based rationale
Q1	1	Fragmentation/concurrent inference question explicit
Q2	1	MIG profiles, merge/split and placement algorithms described

QA	Score	Source-based rationale
Q3	0.5	GPU/memory/CUDA/driver/DCGM clear; OS/backend unresolved
Q4	1	SLMs, 50 jobs, batch ranges and local10-job strata explicit
Q5	0.5	MISO pre-profiled and adapted; ORACLE privileged information disclosed
Q6	1	JCT/makespan/normalized STP distinct; avoid p99 interpretation
Q7	0.5	Prediction perturbation and power-cap sensitivities; repetitions/variance unresolved
Q8	0.5	Table2 and numeric prose readable; plot coordinates missing
Q9	0.5	Reconfiguration cost/architecture and prediction limits discussed; incomplete threats treatment
Q10	0.5	Implementation described but missing runtime/control detail restricts replication

A.20 GA-S020: iGniter: Interference-Aware GPU Resource Provisioning for Predictable DNN Inference in the Cloud

Preferred report: IEEE Transactions on Parallel and Distributed Systems, 2023; report GA-PX005. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	MPS active-thread quotas + interference-aware batch/provisioning and shadow process
Environment	Ten EC2 p3.2xlarge instances, one V100 each, 8vCPU/61GB; Triton2.12.0, TensorRT8.0.1.6. MPS active-thread percentage; allocation unit2.5%. Heterogeneous extension adds20 T4 instances. OS/driver/CUDA version NOT EXTRACTED.
Workload	AlexNet, ResNet50, VGG19(ImageNet), SSD(VOC2012); 12 workload/SLO configurations; constant arrivals. Local experiment co-locates four models at25% quota, batch3. p99 provisioning measured over windows e.g.30s; model-validation experiments repeated3 times with SD bars.
Comparator	FFD+; GSLICE+ with iGniter placement; gpu-lets+ modified batch strategy and best-fit placement. These are adapted baselines.
Bounded finding	Local four-model latency prediction error1.53%-5.02%(Fig13). Fig14 deployment: iGniter6 GPUs/0 SLO violations vs gpu-lets+8/3, FFD+5/10, GSLICE+6/3; 25% lower hourly instance cost vs gpu-lets+, not25% higher local utilization. Fig17 shows an initial p99 violation before shadow-process recovery.

Extraction field	Recorded observation
Transfer limit	Constant arrivals and four vision models limit bursty/LLM transfer. Negligible PCIe contention assumption can underpredict AlexNet; baselines modified. Steady-state/post-correction SLO success is not an unconditional guarantee; shadow standby incurs capacity/memory costs. Chart bars/latency traces unavailable in Markdown.
Source anchor	Published pp820-824; sections4.2, 5.1-5.4; Figs13-18; Markdown lines938-1245

QA	Score	Source-based rationale
Q1	1	Inference interference and SLO provisioning explicit
Q2	1	Active-thread quotas, batching, placement and shadow recovery described
Q3	0.5	EC2 GPU/CPU/memory and serving versions clear; OS/driver missing
Q4	1	Models, datasets, constant arrivals, quotas and local batch3 clear
Q5	0.5	Three adapted baselines and modifications disclosed; effect confounded with placement/batching
Q6	1	p99, SLO throughput, cost and prediction error distinguished
Q7	1	Model validation three repetitions/SD; p99 windows and update rule explicit; not all experiments have same controls
Q8	0.5	Figure captions/numeric prose survive; error bars and full traces not digitizable
Q9	1	PCIe assumption, model error and recovery described
Q10	1	Public code URL, algorithm and coefficients aid interpretation; code not rerun

A.21 GA-S021: InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems

Preferred report: Future Generation Computer Systems, 2024; report GA-PX006. **Current QA:** 7.5/10 (moderate).

Extraction field	Recorded observation
Mechanism	Temporal SLO-aware scheduling + feedback goodput balancing; MPS comparator
Environment	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace. OS/driver/CUDA/backend versions NOT EXTRACTED.

Extraction field	Recorded observation
Workload	Local GoogLeNet/ResNet152 Poisson arrivals; three-model test adds Mobile_Pose_MobileNetv3; 600req/s distributed in six ratios; SLO100ms and interval50ms. Separate61-model, 90min MAF replay on four GPUs.
Comparator	Clockwork; MPS plus Clockwork SLO scheduling with two worker processes/GPU; EDF without batching (not a like-for-like batching baseline).
Bounded finding	Local Table3 load3:1:2: unfairness1.05 vs Clockwork1.50; SLO goodput524.29 vs457.24req/s. Other load ratios give smaller differences. Local unfairness is max/min model slowdown(Eq2), not p99 or security. Four-GPU MAF outcome kept separate.
Transfer limit	Two-column Markdown interleaves algorithms/prose; use labelled Table3 rows and sections, not scrambled formulas. Repetitions/variance and software versions unresolved. MAF trace has minute-level invocation counts; cross-study batching and isolation definitions differ.
Source anchor	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899

QA	Score	Source-based rationale
Q1	1	Heterogeneous model-sharing fairness explicit
Q2	1	Estimator, feedback controller and temporal scheduling described
Q3	0.5	Server/GPU/memory clear; OS/driver/backend unresolved
Q4	1	Models, Poisson load, SLO, interval, load ratios and replay duration explicit
Q5	0.5	Clockwork and adapted MPS; EDF lacks batching and is less directly comparable
Q6	1	Goodput requests/s and unfairness ratio distinguish outcome definitions
Q7	0.5	90min replay and parameter sensitivity clear; local repetitions/variance unresolved
Q8	1	Table3 rows/labels unambiguous in Markdown; plot coordinates absent
Q9	0.5	Future work/memory and heterogeneity discussed; incomplete validity treatment
Q10	0.5	Algorithms interpretable despite column interleaving; software/control gaps restrict replication

A.22 GA-S022: MIGER: Integrating Multi-Instance GPU and Multi-Process Service for Deep Learning Clusters

Preferred report: Proceedings of ICPP 2024, 2024; report GA-PN00001. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	MIG partitions with per-partition MPS online/offline co-location, local partition planner and fragmentation control
Environment	Two servers, each 2 Intel Xeon Gold 5320 CPUs and 4 A100 80GB GPUs; CUDA 12.0; per-partition MPS daemons and DCGM. OS/driver not yet extracted.
Workload	Table 3 inference/training models, including ResNet50, MobileNet, BERT and U-net. 150 jobs of 30 min to 2 h; Poisson arrival parameter 60 s. Local MPS/MIG examples separated from eight-GPU testbed and 128-GPU simulation.
Comparator	MuxFlow (MPS) and modified MISO (MIG); MISO adapted to preserve online jobs during reconfiguration.
Bounded finding	Section 3.1/Fig. 2 directly compares local MPS 57/29/14 percent and MIG 4g/2g/1g allocations. Section 5.2/Fig. 8 reports testbed mean JCT 46.6% below MuxFlow and 36% below MISO; these are job/cluster outcomes, not inference p99 gains. Fig. 9 notes slower small jobs under MIGER.
Transfer limit	Includes mixed training/inference and multi-GPU optimization; retain only explicitly local sharing and inference-QoS evidence for assigned comparison. Abstract and introduction swap comparator order for the two JCT percentages; use section 5.2 mapping. No chart coordinates reconstructed or universal p99 claim.
Source anchor	Printed pp. 505-506, 510-511; sections 3.1-3.3, 5.1-5.4; Tables 1-3; Figs. 2, 4, 8-11; N00001.md lines 119-229, 478-662

QA	Score	Source-based rationale
Q1	1	Online inference versus offline training question explicit (sections 1, 3)
Q2	1	Hybrid shares and partition-ID control described (sections 2, 4)
Q3	0.5	GPU/CPU/CUDA specified; OS/driver not yet transcribed (5.1)
Q4	1	Model/batch/QoS table, jobs and arrival parameter stated (Table 3, 5.1)
Q5	0.5	MISO adaptation disclosed; baseline equivalence is qualified (5.1)
Q6	1	JCT/makespan defined; not interchangeable with request tails (2.2, 5.1)
Q7	0.5	Trace sizes and monitoring interval stated; repetitions/uncertainty not established in conversion (5.1, 5.4)

QA	Score	Source-based rationale
Q8	1	Numerical prose and adverse small-job result readable (5.2); plotted coordinates unused
Q9	1	Small-job slowdown and reconfiguration overhead explicitly discussed (5.2-5.4)
Q10	0.5	Architecture/implementation interpretable; full software/control details incomplete (4.6, 5.1)

A.23 GA-S023: Towards Optimal Preemptive GPU Time-Sharing for Edge Model Serving

Preferred report: Proceedings of the 9th International Workshop on Container Technologies and Container Clouds (WoC 2023), 2024; report GA-PN00005. **Current QA:** 7.0/10 (moderate).

Extraction field	Recorded observation
Mechanism	Deft: adaptive CUDA synchronization frequency controlling kernel queues and priority preemption; API interposition prototype
Environment	One NVIDIA A40 GPU; prototype intercepts CUDA calls. CPU/OS/driver/CUDA versions not yet extracted.
Workload	Low-priority RetinaNet versus high-priority ResNet50; ResNet50 arrives every second; RetinaNet changes batch 1 to 30 every 5 s or alternates 720p/2160p inputs.
Comparator	Implemented synchronization strategies inspired by REEF, PipeSwitch and Shepherd; Deft-no-profile ablation. These are adapted policies, not full system reproductions.
Bounded finding	Sections 3.2 and 5 show optimal synchronization depends on batch and image size; the fixed strategy can impose much larger preempting-job JCT inflation after a workload change. Section 5.2 reports Deft switches to synchronizing every kernel for batch 30; earlier batch-1 case uses every 500 kernels. Table 2 cells are interleaved, so no exact violation-rate row is imported.
Transfer limit	Preliminary six-page workshop evaluation with two principal models and one A40. Exhaustive pair profiling does not scale; no cluster/generative-model implementation claimed (section 6). Keep export year 2024 and printed conference/reference year 2023 separately; both inside window.
Source anchor	Printed pp. 15-17; sections 3.2, 4.2, 5.1-5.3 and 6; Table 2 and Fig. 4 discussed in prose; N00005.md lines 250-468

QA	Score	Source-based rationale
Q1	1	Two priority inference jobs and preemption trade-off explicit (3.1)
Q2	1	Synchronization strategy and predictor described (4.1-4.2)

QA	Score	Source-based rationale
Q3	0.5	A40 identified, remaining software/hardware profile incomplete (5.1)
Q4	1	Models, arrivals, batch/input changes specified (3.2, 5.2-5.3)
Q5	0.5	Baselines are implemented synchronization analogues, limiting fidelity (5.1)
Q6	1	JCT inflation and SLO-violation objectives defined (3.1, 4.2)
Q7	0	Run duration, independent repetitions and uncertainty not established in supplied evaluation (5)
Q8	0.5	Result prose supports direction; Table 2 and chart labels are damaged, limiting numerical verification
Q9	1	Explicit profiling scalability and generative/multi-GPU limitations (6)
Q10	0.5	Concrete prototype and scenarios interpretable; exhaustive profile/data details incomplete (4.2, 6)

A.24 GA-S024: FaST-GShare: Enabling Efficient Spatio-Temporal GPU Sharing in Serverless Computing for Deep Learning Inference

Preferred report: Proceedings of ICPP 2023, 2023; report GA-PN00015. **Current QA:** 7.5/10 (moderate).

Extraction field	Recorded observation
Mechanism	FaST-GShare: MPS spatial shares plus CUDA interception/token-based temporal quotas; CUDA IPC model sharing
Environment	GCP N1 VMs: 48-core/60GB master; four 24-core worker VMs, each V100 16GB (80 SMs). Ubuntu 20.04.5, Kubernetes 1.24.2, OpenFaaS faas-netes 0.14.2/gateway 0.21.3; PyTorch/TensorFlow; CUDA/driver not yet extracted.
Workload	ResNet, RNNT, BERT and GNMT functions; local time shares 20/40/60/80/100 percent and SM shares 6/12/24/50/60/80/100 percent; multi-pod single-GPU profiling distinct from four-worker placement.
Comparator	Time sharing without spatial shares (racing) versus 12%/24% SM allocation; section 5.3 separately tests oversubscribed and non-oversubscribed pods.
Bounded finding	Local section 5.3 states eight RNNT pods at 12% SM shares reach about 40 req/s with tail latency below 500 ms, versus one racing pod about 12 req/s and above 1250 ms. The local ResNet/RNNT comparison in Fig. 9 reports removing their observed interference with 24% shares. No unlabelled tail value is relabelled p99.
Transfer limit	Model-specific/proportional share effects, IPC memory trust and batching limit transfer. Prose fractional-gain notation is ambiguous (3.15x with raw 296.8/71.37), so no such multiplier imported. Supplied plot coordinates/percentile identity and repetitions not reconstructed.

Extraction field	Recorded observation
Source anchor	Printed pp. 641-643; sections 4, 5.1-5.4; Figs. 8-10; N00015.md lines 523-735

QA	Score	Source-based rationale
Q1	1	FaaS inference sharing problem explicit (1)
Q2	1	MPS, quota/token and IPC design described (3-4)
Q3	0.5	VM/GPU/platform versions specified, CUDA/driver incomplete (5.1)
Q4	1	Model/share/pod experiments specified (5.1-5.3)
Q5	0.5	Racing and spatial controls explicit; oversubscription strata must remain separate (5.3)
Q6	0.5	Units and throughput clear; tail percentile not established from extracted source (5.3)
Q7	0.5	Load tools and experiment scenarios reported, repetitions/variance incomplete (5.1-5.3)
Q8	1	RNNT raw outcomes readable in prose; chart pixels not reconstructed (5.3)
Q9	0.5	Memory contention and resource fragmentation discussed, limited validity treatment (3.4-3.5)
Q10	1	Quota controls and platform implementation critically interpretable (3-5)

A.25 GA-S025: SPLIT: QoS-Aware DNN Inference on Shared GPU via Evenly-Sized Model Splitting

Preferred report: Proceedings of ICPP 2023, 2023; report GA-PN00040. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	SPLIT: genetic offline model-block splitting with token-based greedy temporal preemption
Environment	Jetson Nano; Ubuntu 18.04; ONNX Runtime 1.12.1. CUDA/driver versions not yet extracted.
Workload	YOLOv2, GoogLeNet, ResNet50, VGG19, GPT-2; 1000 requests per scenario, Poisson average intervals 160 to 110 ms across six scenarios; latency target alpha times isolated execution time, alpha 2-20.
Comparator	Clockwork-like sequential policy, PREMA-like preemption and Runtime-aware multi-stream approach (RT-A); these adapted baselines differ in concurrency.

Extraction field	Recorded observation
Bounded finding	Section 5.5 reports violation rate below 10% beyond alpha=4 versus 26% under RT-A in the discussed comparison. For short requests, jitter reductions versus Clockwork/PREMA/RT-A are 55.3/46.8/68.9% at low load and 56.0/50.3/69.3% at high load. Jitter is block-execution standard deviation, not p99 latency. Section 6 reports split overhead and static-graph limits.
Transfer limit	One edge GPU and synthetic Poisson load; not a server-class GPU generalization. Temporal predictability sacrifices spatial parallelism; long-request stability can worsen. Table 3 split overhead and RNN limits qualify short-request improvements.
Source anchor	Printed pp. 611-613; sections 5.1-5.5, 6; Tables 1-3; Figs. 6-7; N00040.md lines 451-629

QA	Score	Source-based rationale
Q1	1	Single-GPU per-request QoS question explicit (1)
Q2	1	Block splitting and greedy preemption specified (3-4)
Q3	0.5	GPU/OS/runtime given, driver/CUDA incomplete (5.1)
Q4	1	Five models, 1000 requests, six load intervals and alpha sweep (5.1-5.2)
Q5	0.5	Adapted baseline policies described; device/implementation portability qualified (5.3)
Q6	1	Violation target and block-execution SD defined (5.2)
Q7	0.5	Scenario request count/load explicit, repetitions/uncertainty not established (5)
Q8	1	Outcome prose and split-overhead table readable (5.4-5.5, Table 3)
Q9	1	Overhead, static graphs and adverse longer-request stability discussed (5.5-6)
Q10	1	Mechanism and experiment are critically interpretable despite missing configuration controls (3-6)

A.26 GA-S026: MIGServe: Layout-Aware Multi-Instance GPU Management for Efficient LLM Serving

Preferred report: Proceedings of ICPP 2026, 2026; report GA-PN00831. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	MIGServe: buddy-aware MIG placement, proactive pair migration and temperature-guided eviction

Extraction field	Recorded observation
Environment	One server with four A100 40GB GPUs, two AMD EPYC 7413 CPUs and 512GB RAM; each GPU up to seven MIG slices; OS/driver/serving-engine versions not yet extracted.
Workload	Llama and Qwen 3B-14B models and distinct replicas; Azure/Arena fitted Gamma arrivals, ShareGPT/GSM8K inputs, 4K samples each; request rates 2-12 RPS; TTFT SLO 6 s and TPOT 100 ms at 90% attainment.
Comparator	Static profile distribution and layout-agnostic dynamic best-fit MIG baseline (LA-Dyn), not every feature of cited production schedulers.
Bounded finding	Section 5.2 Azure: sustainable rate at 90% SLO attainment is 9.84 RPS versus LA-Dyn 4.13 and Static 1.85. Arena approximately 10.99 versus 2.37 and 2.48 RPS. These are four-GPU server outcomes; local mechanism is MIG layout control, not a single-GPU throughput multiplier.
Transfer limit	Four-GPU evaluation and constructed layout-agnostic comparator; 90% attainment is weaker than 99% SLO targets used elsewhere. MIG hardware isolation is described, without adversarial/fault experiment. Source pixels and unreported repeats/versions are not reconstructed.
Source anchor	Printed pp. 850-853; sections 4, 5.1-5.4; Fig. 9 and Figs. 11-13; N00831.md lines 519-780

QA	Score	Source-based rationale
Q1	1	Heterogeneous LLM layout/fragmentation problem explicit (1)
Q2	1	Buddy placement, migration and eviction mechanisms described (3-4)
Q3	0.5	Server/GPU/memory given; software versions not yet extracted (5.1)
Q4	1	Models, datasets, arrival construction and SLOs stated (5.1)
Q5	0.5	Two operationally defined baselines; LA-Dyn is a constructed heuristic comparator (5.1)
Q6	1	TTFT/TPOT, attainment and sustainable request rate defined (5.1)
Q7	0.5	Load/SLO sweeps and ablations present; repetitions/variance unresolved (5)
Q8	1	Rates and ablation outcomes readable in prose (5.2-5.4)
Q9	0.5	Layout/reconfiguration trade-offs discussed, limited threat treatment (4-6)
Q10	1	Control logic and workload construction support critical interpretation (3-5)

A.27 GA-S027: MMK: A Hybrid Scheduling Framework for Fine-Grained GPU Sharing for Deep Learning Applications

Preferred report: ACM Transactions on Architecture and Code Optimization, 2026; report GA-PN00839. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	MMK: MIG, elastic MPS oversubscription and CUDA-intercepted kernel-priority scheduling
Environment	Two Xeon Platinum 8369B CPUs and two A100 PCIe 80GB GPUs; Ubuntu 20.04/kernel 5.15; driver 535.154.05, CUDA 12.2, PyTorch 2.4.1, cuDNN 8.9.7, cuBLAS 12.2.
Workload	60 online and 40 offline jobs on testbed, Poisson arrival parameter 30 s; Table 3 CNNs and Qwen2-7B/Llama-3.2-3B/DeepSeek-R1-1.5B; 1000-job simulation is separate.
Comparator	Default time sharing, MPS, MISO and MIGER; fixed 7g.80gb baseline for component analysis.
Bounded finding	Section 4.3/Fig. 9 reports mean JCT 28% lower, makespan 32% lower and normalized progress-rate STP 35% higher than MIGER. These are mixed-job/testbed metrics, not per-request p99 latency. Section 4.5 reports 1.5% MIG and 3.2% MPS reconfiguration lifecycle overhead.
Transfer limit	Mixed inference/training and two-GPU optimization; local priority scheduling is the transferable mechanism. Many figure labels contain NUL/glyph corruption, so numerical prose only. STP/JCT does not establish online tail-latency or fault containment.
Source anchor	Article pp. 84:15-84:19; sections 4.1-4.5; Table 3; Figs. 9-12; N00839.md lines 789-1040

QA	Score	Source-based rationale
Q1	1	Online/offline GPU contention and priorities explicit (1)
Q2	1	Three-level local scheduling and CUDA interposition detailed (3-4.1)
Q3	1	CPU/GPU/kernel/driver/CUDA/framework versions supplied (4.1-4.2)
Q4	1	Job counts, arrival parameter and inference/training table explicit (4.2, Table 3)
Q5	0.5	Baselines and their scheduling rules described; comparator fidelity qualified (4.2)
Q6	1	JCT, STP and makespan defined (4.2)
Q7	0.5	Testbed/simulation sizes stated; repeats/variance not verified (4.2)
Q8	0.5	Numerical prose readable, figures substantially corrupted in conversion (4.3-4.5)
Q9	0.5	Overhead/resource contention discussed, limited separate validity treatment

QA	Score	Source-based rationale
Q10	1	Detailed stack/control permits critical interpretation; experiments not rerun (4.1-4.5)

A.28 GA-S028: BOER: Enhancing Resource Utilization for Deep Learning Inference with Hybrid Spatial GPU Sharing

Preferred report: Proceedings of SC 2025, 2025; report GA-PN00844. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	BOER: MIG model grouping with Bayesian-optimized MPS configurations inside partitions
Environment	Two servers, each two Xeon Gold 5320 CPUs and four A100 80GB GPUs; CUDA 12.6. Single-A100 motivation profiling separated from eight-GPU system evaluation.
Workload	Nine inference models across five multi-model scenarios; Poisson arrivals; each application request fans out to each model; selected batching based on demand and model QoS.
Comparator	gpulet; USHER IR-Scheduler only; PARIS/ELSA; MIG-serving fast algorithm only; ParvaGPU. Full omitted features are not benchmarked.
Bounded finding	Local section 2.4/Fig. 3 observes ParvaGPU can underperform MIG-only serving for larger BERT/ResNet50 while helping lightweight MobileNet. Section 5.2/Fig. 8 gives eight-GPU QoS-preserved throughput mean 34.61% above ParvaGPU; 56.8/62.49/34.63/44.83% above MIG-serving/gpulet/USHER/PARIS. Keep local qualitative counterexample separate from fleet improvements.
Transfer limit	Bayesian exploration cost and instance interference qualify hybrid gains. Baseline reductions (MIG-serving fast/USHER IR only) and multi-GPU request fan-out constrain comparison. Same senior author as MIGER, but different inference-only system and experiments; not a same-study report extension identified in supplied text.
Source anchor	Printed pp. 521, 526-528; sections 2.4, 4.6, 5.1-5.4; Tables 2-3; Figs. 3, 8, 10; N00844.md lines 200-265, 700-890

QA	Score	Source-based rationale
Q1	1	QoS-aware multi-model hybrid sharing question explicit (1-2)
Q2	1	Grouping, partition selection and MPS Bayesian search specified (4)
Q3	0.5	GPU/CPU/CUDA clear, OS/driver not yet extracted (5.1)
Q4	1	Models, input/QoS tables and fan-out/arrival design stated (5.1)
Q5	0.5	Fast-only MIG-serving and IR-only USHER restrictions disclosed (5.1)

QA	Score	Source-based rationale
Q6	1	QoS-preserved RPS measured by raising rate to violation (5.2)
Q7	0.5	Sweeps/scenarios/ablations present, repetitions and confidence unresolved (5)
Q8	1	Explicit normalized-result prose and local counterexample readable (2.4, 5.2)
Q9	1	Intra-partition interference, search cost and reconfiguration discussed (2.4, 4.6)
Q10	1	Algorithms/open prototype and bounded baseline choices interpretable (4-5)

A.29 GA-S029: Adaptive GPU Sharing for Real-time LLM Serving with Best-effort Workloads

Preferred report: ICPP Workshops 2026, 2026; report GA-PN00872. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	AdMix: adaptive MPS active-thread caps, lightweight LLM estimator and best-effort pause/full-access controls
Environment	AMD EPYC 7713 and two A100 PCIe 80GB GPUs with NVLink; CUDA 12.3; vLLM 0.10.2. LLaMA-7B uses TP1 (local), Qwen-14B TP2 is separate.
Workload	LlaMA-7B/Qwen-14B with best-effort ResNet50/VGG19/UNet3D/DeepLabV3 inference; Azure/BurstGPT five-hour segments compressed to one hour and load-scaled. TTFT 1s, TPOT 200ms; MPS 5% steps and 10s switch interval.
Comparator	Unpartitioned driver time sharing and static MPS selected by exhaustive target-trace replay to meet 99% attainment; batch choices differ between temporal and static MPS controls.
Bounded finding	Section 4.2 reports average best-effort throughput 1.58x Static MPS with local LLaMA-7B (TP1) and 3.9x with Qwen-14B (TP2), keeping strata distinct. AdMix still increases mean TTFT/TPOT versus LLM-only; on Qwen its mean latency increases exceed Static MPS. More throughput is not zero interference.
Transfer limit	Full six-page original workshop report; one/two-GPU workload strata and tuned-on-test-trace static reference limit transfer. Estimates of average latency do not establish p99. No testbed code rerun or security/fault experiment inferred.
Source anchor	Printed pp. 3-5; sections 3 and 4.1-4.2; Table 1; Figs. 5-6; N00872.md lines 270-453

QA	Score	Source-based rationale
Q1	1	Real-time LLM and best-effort inference problem explicit (1)
Q2	1	MPS controller, proactive pause and full-access path described (3)
Q3	0.5	Hardware/CUDA/engine versions stated, OS/driver incomplete (4.1)
Q4	1	Models, trace compression, quotas and SLOs specified (4.1, Table 1)
Q5	0.5	Oracle-like static trace search and differing temporal batches disclosed (4.1)
Q6	1	Mean TTFT/TPOT, attainment and normalized BE throughput defined (4.1)
Q7	0.5	One-hour trace and load scaling explicit; independent repeats/variance unresolved (4.1)
Q8	1	Local TP1 and separate TP2 results readable in prose (4.2)
Q9	1	Static baseline impracticality and latency/throughput cost discussed (4.1-4.2)
Q10	1	Control parameters and implementation hooks interpretable (3-4)

A.30 GA-S030: SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs

Preferred report: Proceedings of PPoPP 2025, 2025; report GA-PN00904. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	SGDRC: CUDA/TVM runtime with software VRAM-channel coloring, bimodal tensors and tidal SM masks
Environment	Separate Tesla P40 and RTX A2000 evaluations; clients/server on same machine. TVM-generated kernels and shared CUDA context. OS/driver/CUDA/CPU profile not yet fully extracted.
Workload	Apollo trace for latency-sensitive inference, closed-loop best-effort DNNs; four instances per LS model, no dynamic batching. SLO = n times model isolated p99 runtime for n concurrent DNN services; light/heavy loads.
Comparator	TGS, MPS where supported, Orion, multi-streaming and SGDRC-static; Fig. 18 explicitly excludes unsupported P40 MPS.
Bounded finding	Section 9.3: versus Orion, P40 overall throughput up to 1.47x and BE up to 2.36x; A2000 up to 1.32x and 1.79x, respectively. Mean SLO attainment 99.0% across evaluated settings. Section 10 explicitly states shared-context GPU runtime errors are not isolated.

Extraction field	Recorded observation
Transfer limit	Software coloring is not a proven security boundary. MIG integration is future work; not benchmarked on A100/H100. Reverse-engineering/calibration and TVM compatibility matter; do not treat throughput maxima as averages or infer a P40 MPS result.
Source anchor	Printed pp. 276-277; sections 9.2-9.3, 10; Table 3 and Fig. 18; N00904.md lines 863-965

QA	Score	Source-based rationale
Q1	1	Concurrent LS/BE inference and hardware allocation problem explicit (1)
Q2	1	Coloring, tensor conversion and dynamic SM allocation detailed (5-8)
Q3	0.5	P40/A2000 and TVM stated; remaining environment details not yet transcribed (9.2)
Q4	1	Trace replay, BE loop, instance count and batching policy explicit (9.2)
Q5	0.5	Unsupported P40 MPS omitted and static ablation included; kernel/runtime comparability qualified (Fig.18)
Q6	1	p99, n-scaled SLO, goodput/throughput units defined (9.2)
Q7	0.5	Light/heavy trace regimes stated; repetitions/uncertainty not established from extracted source (9)
Q8	1	Separate GPU maxima and SLO outcomes readable in prose (9.3)
Q9	1	Shared-context fault failure and future MIG portability explicitly acknowledged (10)
Q10	1	Architecture/methods allow critical interpretation, despite environment gaps (5-10)

A.31 GA-S031: Beyond Utilization: Energy-Conscious GPU Sharing for Inference Serving

Preferred report: Proceedings of SOSP 2026, 2026; report GA-PN00946. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	EnerTune: MIG allocation with joint frequency/batch selection, colocation-power model and runtime tuner; separate MPS portability experiments
Environment	16 A100 80GB GPUs (four/server), AMD EPYC 9554P; Ubuntu 22.04, driver 575.57.08, CUDA 12.9. Eight A6000/RTX4090 portability experiments separate.

Extraction field	Recorded observation
Workload	22 instances of 11 CNN, SDXL, Whisper and GPT-J-6B models; dense/bursty MAF traces, about one hour, 25-125% load; p99 normalized by per-model SLO derived from isolated Poisson p90.
Comparator	Usher, FGD, GPULets, ParvaGPU; additive-power estimator ablation. Local pair-energy examples compared with isolated execution.
Bounded finding	Section 2.1/Fig. 2: co-location choice can increase energy; SDXL plus MobileNetV3 example uses 13% less energy versus no sharing despite 1.14x latency. Section 7.1/Fig. 9: 16-GPU testbed energy reduced by factors 1.4-2.3 and power by 1.3-2.6 versus baselines while meeting SLOs; fleet results are not local per-GPU savings.
Transfer limit	Joint power/energy objectives select different allocations. Local model-pair evidence must stay distinct from fleet savings and from frequency-only effects. Source chart pixels not available; experiments/code not rerun; missing warm-up and independent-run uncertainty not assumed.
Source anchor	Printed pp. 66-67, 73-75; sections 2.1-2.2, 6, 7.1-7.2; Figs. 2, 9; N00946.md lines 140-249, 798-995

QA	Score	Source-based rationale
Q1	1	Inference multiplexing with power/energy objective explicit (1)
Q2	1	MIG allocation, energy models and runtime frequency/batch tuning specified (3-6)
Q3	1	GPU/server/OS/driver/CUDA supplied (7)
Q4	1	Models, one-hour traces, loads and SLO derivation explicit (7)
Q5	0.5	Same workloads and representative comparators; cross-system design differences qualified (7)
Q6	1	Energy versus power and normalized p99/SLO metrics separated (7)
Q7	0.5	Trace duration/load sweep present; warm-up/repetition uncertainty unresolved (7)
Q8	1	Bounded pair outcomes and system gain prose readable (2.1, 7.1)
Q9	1	Energy-incompatible pairs and power-versus-energy allocation conflict discussed (2, 7)
Q10	1	Stack, models, equations and implementation support critical interpretation (3-7)

A.32 GA-S032: FluidFaaS: A Dynamic Pipelined Solution for Serverless Computing with Strong Isolation-based GPU Sharing

Preferred report: Proceedings of HPDC 2025, 2025; report GA-PN00920. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	MIG partitioned serverless functions with pipelined instance construction, eviction and dynamic resource assignment
Environment	A100 80GB MIG profiles described; evaluation spans serverless deployment and GPU allocations. Per-GPU comparisons separated from aggregate system metrics; full driver/runtime settings not extracted.
Workload	Image classification and depth recognition DNN function pipelines, light/medium/heavy invocation traces derived from Azure Functions; SLO set to 1.5 times isolated latency in default setup.
Comparator	ESG MIG serverless baseline and INFless; baseline variants differ in pipeline execution.
Bounded finding	Section 7.1/Fig. 9 reports FluidFaaS SLO hit-rate improvement of 90% for medium and 61% for heavy workload mixes over the compared systems; the abstract reports 25–75% throughput increases across workloads. These are system-level results, not a measured per-GPU utilization effect.
Transfer limit	Pipeline, placement and MIG allocation contribute jointly; comparisons with a non-pipelined baseline cannot isolate the benefit of MIG partitioning alone. No adversarial security or fault-injection evidence is inferred from MIG descriptions.
Source anchor	N00920.md abstract, sections 3, 6, 7.1; Tables 2, 5–6; Figs. 9 and 12

QA	Score	Source-based rationale
Q1	1	Serverless multi-function tenant question explicit (intro)
Q2	1	MIG partition and pipeline controls described (sections 3–5)
Q3	0.5	A100/profile specified, full software stack not extracted (section 6)
Q4	1	Three load regimes, functions, traces and SLO scale (section 6)
Q5	0.5	ESG and INFless compared but pipeline semantics differ (section 6)
Q6	1	SLO hit and latency/throughput metrics described (sections 6–7)
Q7	0.5	Trace derivation stated; repetitions and variance incomplete in conversion (section 6)
Q8	1	Tables 5–6 and Figs. 9, 12 support bounded claims (section 7)
Q9	1	Fragmentation and baseline trade-offs discussed (sections 3, 7)
Q10	0.5	Enough setup for critical interpretation; full implementation settings limited (sections 6–7)

A.33 GA-S033: Improving GPU Sharing Performance through Adaptive Bubbleless Spatial-Temporal Sharing

Preferred report: Proceedings of EuroSys 2025, 2025; report GA-PN01085. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	Bless: adaptive kernel-squad scheduling, MPS contexts with SM affinity and temporal/spatial quota adjustments
Environment	One A100 40GB (108 SM), AMD EPYC 7302, CUDA 11.8, Ubuntu 20.04 (section 6.1).
Workload	Co-located inference and training DNNs; explicit two- and four-instance BERT inference scenarios and VGG11/ResNet50 quotas on one GPU.
Comparator	Temporal round robin and quota-based spatial sharing/MPS configurations in section 6; relative percentages must retain their stated comparator and scenario.
Bounded finding	The introduction and evaluation report mean latency reductions of 39.1% for two BERT inference instances and 41.2% for four, under the paper's compared GPU-sharing systems. Other reported aggregate gains are not recast as p99 or universal speedups.
Transfer limit	Quota, model pairing and GPU generation condition results. The conversion has damaged column/table layout, so no unlabeled plotted coordinates or precise tail-percentile gains were recovered.
Source anchor	N01085.md intro, sections 4, 6.1–6; Fig. 1 and BERT co-location result prose

QA	Score	Source-based rationale
Q1	1	Multiple inference tenant question explicit (intro)
Q2	1	Kernel squads and affinity/quota controls described (sections 4–5)
Q3	1	Single A100/CPU/OS/CUDA specified (6.1)
Q4	1	DNN model pairs and quota scenarios specified (Fig. 1, 6)
Q5	0.5	Multiple sharing baselines; exact equal resource budgets partly unclear in conversion (6)
Q6	1	Latency and quota metrics identifiable; mean is not p99 (6)
Q7	0.5	Trace and setup given; variance/repetitions not fully recovered (6.1)
Q8	1	BERT result in prose and figures support scoped comparisons (6)
Q9	0.5	Some adverse cases/transfer limits discussed, no security/fault test (6)
Q10	0.5	Mechanism interpretable; damaged plot/table layout limits reconstruction (4–6)

A.34 GA-S034: Understanding GPU Resource Interference One Level Deeper

Preferred report: Proceedings of SoCC 2025, 2025; report GA-PN01096. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	CUDA streams and CUDA Green Context SM partition for controlled co-location; interference characterization rather than a deployed scheduler
Environment	H100 NVL (132 SM, CUDA 12.9, driver 575.57.08) and RTX 3090 (82 SM, CUDA 12.6, driver 560.35.03); modified vLLM fork.
Workload	Llama 3.1 8B and Gemma 3 1B decode phases co-located with controlled sleep, copy or compute kernels; Table 1 H100 batch 8, 16,384-token prompt, ten generated tokens.
Comparator	LLM decode alone versus controlled co-location in streams/Green Contexts; synthetic interference kernels, not a second production inference service.
Bounded finding	H100 Llama decode P90 time between tokens rises from 7.53 to 16.56 ms under a sleep co-tenant in section 4.2 (batch 1, 1,000-token prompt). Table 1's Green Context experiment isolates SM sets (64/68) but reports P90 TBT as high as 18.38 ms under copy-kernel memory pressure; its alone baseline is 14.2 ms on the whole GPU, so partition cost and interference cannot be collapsed into one effect.
Transfer limit	Synthetic co-tenants and short decode traces quantify resource contention; the paper proposes a future scheduler but does not implement an end-to-end serving controller. Green Context SM partition does not partition L2 or external memory bandwidth. Table 1 conversion is interleaved; only prose and clearly attributable values are used.
Source anchor	N01096.md sections 3–5, especially 4.1–4.4; Fig. 2 and Table 1

QA	Score	Source-based rationale
Q1	1	Interference for concurrent inference explicit (abstract, 4)
Q2	1	Streams and Green Context partition specified (3–4)
Q3	1	GPU/SM/CUDA/driver and modified vLLM stated (4.1)
Q4	0.5	LLM sizes/prompt/batch given; co-tenant synthetic rather than service trace (4.1–4.4)
Q5	0.5	Solo baseline exists; whole-GPU and partition budgets differ (Table 1)
Q6	1	P90 TBT and time units stated (4.2–4.3)
Q7	0.5	Short traces and controlled kernels; repetition/variance limited (4)
Q8	1	Trace/Fig. 2 and Table 1/prose substantiate scoped findings (4)

QA	Score	Source-based rationale
Q9	1	Shared cache/bandwidth and scheduler outlook explicitly discussed (4–5)
Q10	0.5	Benchmarks interpretable; no complete deployed scheduler to reproduce (5)

A.35 GA-S035: Reaching for the Sky: Maximizing Deep Learning Inference Throughput on Edge Devices with AI Multi-Tenancy

Preferred report: ACM Transactions on Internet Technology, 2023; report GA-PN00027. **Current QA:** 7.0/10 (moderate).

Extraction field	Recorded observation
Mechanism	Concurrent model execution and dynamic placement across edge device GPU/CPU and Edge TPU
Environment	Jetson Xavier NX and other edge devices; heterogeneous accelerator comparisons
Workload	Image classification models with concurrent inference on an edge device
Comparator	Single-model or alternate edge execution arrangements
Bounded finding	Abstract and multi-tenancy evaluation report inference throughput gains under concurrent execution; do not attribute heterogeneous placement gains solely to GPU sharing.
Transfer limit	Edge GPU, CPU and TPU contributions are coupled; throughput is not latency predictability or security isolation.
Source anchor	N00027.md abstract; multi-tenancy and performance evaluation sections, figures on CME and DMP

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00027.md abstract; multi-tenancy and performance evaluation sections, figures on CME and DMP
Q2	1	Mechanism described: Concurrent model execution and dynamic placement across edge device GPU/CPU and Edge TPU
Q3	0.5	Hardware/software specified partly: Jetson Xavier NX and other edge devices; heterogeneous accelerator comparisons
Q4	0.5	Workload described: Image classification models with concurrent inference on an edge device
Q5	0.5	Baselines described but equivalence may vary: Single-model or alternate edge execution arrangements

QA	Score	Source-based rationale
Q6	1	Outcome metric located: Abstract and multi-tenancy evaluation report inference throughput gains under concurrent execution; do not attribute heterogeneous placement gains solely to GPU sharing.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00027.md abstract; multi-tenancy and performance evaluation sections, figures on CME and DMP
Q8	1	Bounded result can be traced: N00027.md abstract; multi-tenancy and performance evaluation sections, figures on CME and DMP
Q9	0.5	Alternative explanation or limit recorded: Edge GPU, CPU and TPU contributions are coupled; throughput is not latency predictability or security isolation.
Q10	0.5	Partial configuration and source conversion limit reproduction: Jetson Xavier NX and other edge devices; heterogeneous accelerator comparisons

A.36 GA-S036: Miriam: Exploiting Elastic Kernels for Real-time Multi-DNN Inference on Edge GPU

Preferred report: Proceedings of ACM/IEEE Symposium on Edge Computing 2023, 2023; report GA-PN00046. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	Elastic kernels and contention-aware coordination for mixed-criticality inference
Environment	Edge GPUs, which the paper says lack server MIG/MPS support
Workload	Concurrent critical and best-effort DNN inference tasks
Comparator	Edge multi-DNN scheduling baselines
Bounded finding	Abstract reports up to 92% throughput improvement while critical tasks incur less than 10% latency overhead in its evaluated scenarios.
Transfer limit	Edge elastic-kernel scheduling is transferable temporal/spatial evidence, not a direct MIG/MPS comparison.
Source anchor	N00046.md abstract; sections on kernel coordination and evaluation

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00046.md abstract; sections on kernel coordination and evaluation
Q2	1	Mechanism described: Elastic kernels and contention-aware coordination for mixed-criticality inference
Q3	0.5	Hardware/software specified partly: Edge GPUs, which the paper says lack server MIG/MPS support

QA	Score	Source-based rationale
Q4	1	Workload described: Concurrent critical and best-effort DNN inference tasks
Q5	0.5	Baselines described but equivalence may vary: Edge multi-DNN scheduling baselines
Q6	1	Outcome metric located: Abstract reports up to 92% throughput improvement while critical tasks incur less than 10% latency overhead in its evaluated scenarios.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00046.md abstract; sections on kernel coordination and evaluation
Q8	1	Bounded result can be traced: N00046.md abstract; sections on kernel coordination and evaluation
Q9	1	Source describes scenario-dependent limitations, recorded here: Edge elastic-kernel scheduling is transferable temporal/spatial evidence, not a direct MIG/MPS comparison.
Q10	0.5	Partial configuration and source conversion limit reproduction: Edge GPUs, which the paper says lack server MIG/MPS support

A.37 GA-S037: ElasticRoom: Multi-Tenant DNN Inference Engine via Co-design with Resource-constrained Compilation and Strong Priority Scheduling

Preferred report: Proceedings of HPDC 2024, 2024; report GA-PN00048. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	Resource-constrained compilation and priority scheduling with concurrent GPU inference
Environment	GPU inference testbed; engine-level compilation and scheduling jointly changed
Workload	Concurrent real-time and best-effort DNN requests
Comparator	REEF, GSLICE and other evaluated inference-serving baselines
Bounded finding	The evaluated engine balances request goodput, throughput and latency under co-located inference; comparison is engine-level.
Transfer limit	Compiler and priority scheduling effects cannot be isolated as pure partition effects; plotted numbers were not transcribed.
Source anchor	N00048.md abstract; section 6.2 evaluation and Fig. 1

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00048.md abstract; section 6.2 evaluation and Fig. 1

QA	Score	Source-based rationale
Q2	1	Mechanism described: Resource-constrained compilation and priority scheduling with concurrent GPU inference
Q3	0.5	Hardware/software specified partly: GPU inference testbed; engine-level compilation and scheduling jointly changed
Q4	1	Workload described: Concurrent real-time and best-effort DNN requests
Q5	1	Explicit within-paper comparators identified, with scenario differences still qualified: REEF, GSLICE and other evaluated inference-serving baselines
Q6	1	Outcome metric located: The evaluated engine balances request goodput, throughput and latency under co-located inference; comparison is engine-level.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00048.md abstract; section 6.2 evaluation and Fig. 1
Q8	1	Bounded result can be traced: N00048.md abstract; section 6.2 evaluation and Fig. 1
Q9	0.5	Alternative explanation or limit recorded: Compiler and priority scheduling effects cannot be isolated as pure partition effects; plotted numbers were not transcribed.
Q10	0.5	Partial configuration and source conversion limit reproduction: GPU inference testbed; engine-level compilation and scheduling jointly changed

A.38 GA-S038: SwitchFlow: Preemptive Multitasking for Deep Learning

Preferred report: Proceedings of Middleware 2021, 2021; report GA-PN00049. **Current QA:** 9.5/10 (high).

Extraction field	Recorded observation
Mechanism	Preemptive subgraph scheduling and interleaved model executors
Environment	GTX 1080 Ti, RTX 2080 Ti, V100 and Jetson TX2; TensorFlow, CUDA 10.0, cuDNN 7.6.4
Workload	Section 5.4 co-runs inference with inference; section 5.2 separately co-runs inference with training
Comparator	Session-based time slicing, threaded TensorFlow and MPS, depending on experiment
Bounded finding	Section 5.4/Fig. 10 reports interleaved execution gains among inference jobs of up to 30% versus session-based time slicing in tested pairings.
Transfer limit	The 19.05x p95 inference latency example in Fig. 6 uses a training co-tenant and is not evidence for two inference tenants.
Source anchor	N00049.md sections 5.1, 5.2.1 and 5.4; Figs. 6 and 10

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00049.md sections 5.1, 5.2.1 and 5.4; Figs. 6 and 10
Q2	1	Mechanism described: Preemptive subgraph scheduling and interleaved model executors
Q3	1	GPU, operating system and runtime configuration reported in source: GTX 1080 Ti, RTX 2080 Ti, V100 and Jetson TX2; TensorFlow, CUDA 10.0, cuDNN 7.6.4
Q4	1	Workload described: Section 5.4 co-runs inference with inference; section 5.2 separately co-runs inference with training
Q5	1	Explicit within-paper comparators identified, with scenario differences still qualified: Session-based time slicing, threaded TensorFlow and MPS, depending on experiment
Q6	1	Outcome metric located: Section 5.4/Fig. 10 reports interleaved execution gains among inference jobs of up to 30% versus session-based time slicing in tested pairings.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00049.md sections 5.1, 5.2.1 and 5.4; Figs. 6 and 10
Q8	1	Bounded result can be traced: N00049.md sections 5.1, 5.2.1 and 5.4; Figs. 6 and 10
Q9	1	Source describes scenario-dependent limitations, recorded here: The 19.05x p95 inference latency example in Fig. 6 uses a training co-tenant and is not evidence for two inference tenants.
Q10	1	Detailed testbed and comparison permit critical interpretation, without independent replication: GTX 1080 Ti, RTX 2080 Ti, V100 and Jetson TX2; TensorFlow, CUDA 10.0, cuDNN 7.6.4

A.39 GA-S039: ITIF: Integrated Transformers Inference Framework for Multiple Tenants on GPU

Preferred report: Proceedings of ICPP 2023, 2023; report GA-PN00059. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	Shared transformer backbone with operator-level multi-tenant execution
Environment	GPU with concurrent transformer inference; model architecture and scheduling jointly optimized
Workload	Four tenants per evaluated model and transformer inference requests
Comparator	Separate multi-tenant GPU inference baselines including MPS or stream execution
Bounded finding	Abstract reports up to 2.40x performance versus baselines for evaluated shared-backbone transformer inference.
Transfer limit	Shared backbone changes model computation; speedup cannot be assigned to GPU scheduling alone or generalized to unrelated models.

Extraction field	Recorded observation
Source anchor	N00059.md abstract; design and experiment sections including four-tenant setup

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00059.md abstract; design and experiment sections including four-tenant setup
Q2	1	Mechanism described: Shared transformer backbone with operator-level multi-tenant execution
Q3	0.5	Hardware/software specified partly: GPU with concurrent transformer inference; model architecture and scheduling jointly optimized
Q4	1	Workload described: Four tenants per evaluated model and transformer inference requests
Q5	0.5	Baselines described but equivalence may vary: Separate multi-tenant GPU inference baselines including MPS or stream execution
Q6	1	Outcome metric located: Abstract reports up to 2.40x performance versus baselines for evaluated shared-backbone transformer inference.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00059.md abstract; design and experiment sections including four-tenant setup
Q8	1	Bounded result can be traced: N00059.md abstract; design and experiment sections including four-tenant setup
Q9	1	Source describes scenario-dependent limitations, recorded here: Shared backbone changes model computation; speedup cannot be assigned to GPU scheduling alone or generalized to unrelated models.
Q10	0.5	Partial configuration and source conversion limit reproduction: GPU with concurrent transformer inference; model architecture and scheduling jointly optimized

A.40 GA-S040: Towards SLO-Compliant and Cost-Effective Serverless Computing on Emerging GPU Architectures

Preferred report: Proceedings of Middleware 2024, 2024; report GA-PN00063. **Current QA:** 8.5/10 (high).

Extraction field	Recorded observation
Mechanism	Protean orchestrates MIG partitioning and MPS sharing for GPU serverless inference
Environment	Eight A100 GPUs; local MIG/MPS allocations embedded in a multi-GPU system
Workload	22 ML inference workloads with trace-driven serverless arrivals
Comparator	Evaluated serverless GPU allocation baselines
Bounded finding	The evaluation reports SLO compliance and cost effects of orchestration under trace-driven inference; no per-GPU improvement is extracted from system-level totals.
Transfer limit	Multiple GPUs and global scheduling contribute jointly; local mechanism claims cannot be isolated without matched per-device comparisons.
Source anchor	N00063.md abstract, sections on MIG/MPS design and evaluation of 22 workloads

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00063.md abstract, sections on MIG/MPS design and evaluation of 22 workloads
Q2	1	Mechanism described: Protean orchestrates MIG partitioning and MPS sharing for GPU serverless inference
Q3	1	GPU, operating system and runtime configuration reported in source: Eight A100 GPUs; local MIG/MPS allocations embedded in a multi-GPU system
Q4	1	Workload described: 22 ML inference workloads with trace-driven serverless arrivals
Q5	0.5	Baselines described but equivalence may vary: Evaluated serverless GPU allocation baselines
Q6	1	Outcome metric located: The evaluation reports SLO compliance and cost effects of orchestration under trace-driven inference; no per-GPU improvement is extracted from system-level totals.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00063.md abstract, sections on MIG/MPS design and evaluation of 22 workloads
Q8	1	Bounded result can be traced: N00063.md abstract, sections on MIG/MPS design and evaluation of 22 workloads

QA	Score	Source-based rationale
Q9	1	Source describes scenario-dependent limitations, recorded here: Multiple GPUs and global scheduling contribute jointly; local mechanism claims cannot be isolated without matched per-device comparisons.
Q10	0.5	Partial configuration and source conversion limit reproduction: Eight A100 GPUs; local MIG/MPS allocations embedded in a multi-GPU system

A.41 GA-S041: WattWiser: Power & Resource-Efficient Scheduling for Multi-Model Multi-GPU Inference Servers

Preferred report: Proceedings of IGSC 2023, 2023; report GA-PN00068. **Current QA:** 9.5/10 (high).

Extraction field	Recorded observation
Mechanism	Sharing-aware placement and request routing to co-located models on physical GPUs
Environment	One server with four AMD MI50 GPUs and ROCm 5.2; no NVIDIA MIG/MPS implementation
Workload	Pairs of concurrent image model inference under varying Facebook SWIM-derived traces
Comparator	Uniform and alternative GPU routing/power scheduling policies
Bounded finding	Section 4/Fig. 3 shows model tail latency relative to SLO while requests consolidate onto shared GPUs; power savings are whole-server effects.
Transfer limit	The four-GPU power outcome cannot estimate per-GPU partition efficiency; MI50 setup is not a MIG evaluation.
Source anchor	N00068.md sections 4.1–4.2 and Fig. 3

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00068.md sections 4.1–4.2 and Fig. 3
Q2	1	Mechanism described: Sharing-aware placement and request routing to co-located models on physical GPUs
Q3	1	GPU, operating system and runtime configuration reported in source: One server with four AMD MI50 GPUs and ROCm 5.2; no NVIDIA MIG/MPS implementation
Q4	1	Workload described: Pairs of concurrent image model inference under varying Facebook SWIM-derived traces
Q5	1	Explicit within-paper comparators identified, with scenario differences still qualified: Uniform and alternative GPU routing/power scheduling policies

QA	Score	Source-based rationale
Q6	1	Outcome metric located: Section 4/Fig. 3 shows model tail latency relative to SLO while requests consolidate onto shared GPUs; power savings are whole-server effects.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00068.md sections 4.1–4.2 and Fig. 3
Q8	1	Bounded result can be traced: N00068.md sections 4.1–4.2 and Fig. 3
Q9	1	Source describes scenario-dependent limitations, recorded here: The four-GPU power outcome cannot estimate per-GPU partition efficiency; MI50 setup is not a MIG evaluation.
Q10	1	Detailed testbed and comparison permit critical interpretation, without independent replication: One server with four AMD MI50 GPUs and ROCm 5.2; no NVIDIA MIG/MPS implementation

A.42 GA-S042: DeInfer: A GPU resource allocation algorithm with spatial sharing for near-deterministic inferring tasks

Preferred report: Proceedings of ICPP 2024, 2024; report GA-PN00070. **Current QA:** 9.0/10 (high).

Extraction field	Recorded observation
Mechanism	Interference-aware MPS spatial allocation with latency prediction
Environment	RTX 3080 GPUs, Ubuntu 20.04, NVIDIA driver 470.199.02; local concurrent MPS configuration
Workload	Co-located ResNet-18 and VGG-19 plus inference model combinations
Comparator	Prediction and allocation baselines as reported in section 5
Bounded finding	Section 5 evaluates prediction accuracy and allocation for concurrent inference with MPS; its Fig. 2 shows that MPS spatial limits do not eliminate shared-resource latency interference.
Transfer limit	Two GPUs are present in testbed; claims here concern within-device workloads, not a two-device pooled speedup.
Source anchor	N00070.md sections 2.2 and 5.1–5.2; Fig. 2, Table 1 and testbed table

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00070.md sections 2.2 and 5.1–5.2; Fig. 2, Table 1 and testbed table
Q2	1	Mechanism described: Interference-aware MPS spatial allocation with latency prediction
Q3	1	GPU, operating system and runtime configuration reported in source: RTX 3080 GPUs, Ubuntu 20.04, NVIDIA driver 470.199.02; local concurrent MPS configuration

QA	Score	Source-based rationale
Q4	1	Workload described: Co-located ResNet-18 and VGG-19 plus inference model combinations
Q5	0.5	Baselines described but equivalence may vary: Prediction and allocation baselines as reported in section 5
Q6	1	Outcome metric located: Section 5 evaluates prediction accuracy and allocation for concurrent inference with MPS; its Fig. 2 shows that MPS spatial limits do not eliminate shared-resource latency interference.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00070.md sections 2.2 and 5.1–5.2; Fig. 2, Table 1 and testbed table
Q8	1	Bounded result can be traced: N00070.md sections 2.2 and 5.1–5.2; Fig. 2, Table 1 and testbed table
Q9	1	Source describes scenario-dependent limitations, recorded here: Two GPUs are present in testbed; claims here concern within-device workloads, not a two-device pooled speedup.
Q10	1	Detailed testbed and comparison permit critical interpretation, without independent replication: RTX 3080 GPUs, Ubuntu 20.04, NVIDIA driver 470.199.02; local concurrent MPS configuration

A.43 GA-S043: ESG: Pipeline-Conscious Efficient Scheduling of DNN Workflows on Serverless Platforms with Shareable GPUs

Preferred report: Proceedings of HPDC 2024, 2024; report GA-PN00073. **Current QA:** 8.0/10 (high).

Extraction field	Recorded observation
Mechanism	Pipeline-aware placement combining GPU sharing, batch size and SLO distribution
Environment	GPU-backed serverless workflows; allocation and scheduling across invokers
Workload	Concurrent DNN inference workflow functions and request traces
Comparator	INFless, Fast-GShare, Orion and Aquatope with adapted GPU sharing support
Bounded finding	Abstract reports 61–80% SLO hit-rate improvement and 47–187% cost savings over prior systems for evaluated workflows.
Transfer limit	Figures are workflow/system results influenced by batching and scheduler search, not isolated effects of local GPU sharing.
Source anchor	N00073.md abstract, sections 3 and 5; evaluation setup and end-to-end comparison

QA	Score	Source-based rationale
Q1	1	Concurrent inference question: N00073.md abstract, sections 3 and 5; evaluation setup and end-to-end comparison
Q2	1	Mechanism described: Pipeline-aware placement combining GPU sharing, batch size and SLO distribution
Q3	0.5	Hardware/software specified partly: GPU-backed serverless workflows; allocation and scheduling across invokers
Q4	1	Workload described: Concurrent DNN inference workflow functions and request traces
Q5	0.5	Baselines described but equivalence may vary: INFless, Fast-GShare, Orion and Aquatope with adapted GPU sharing support
Q6	1	Outcome metric located: Abstract reports 61–80% SLO hit-rate improvement and 47–187% cost savings over prior systems for evaluated workflows.
Q7	0.5	Measurement variance/repetition not established by extracted text: N00073.md abstract, sections 3 and 5; evaluation setup and end-to-end comparison
Q8	1	Bounded result can be traced: N00073.md abstract, sections 3 and 5; evaluation setup and end-to-end comparison
Q9	1	Source describes scenario-dependent limitations, recorded here: Figures are workflow/system results influenced by batching and scheduler search, not isolated effects of local GPU sharing.
Q10	0.5	Partial configuration and source conversion limit reproduction: GPU-backed serverless workflows; allocation and scheduling across invokers

References

[1] B. Zhang, S. Li, and Z. Li, "MIGER: Integrating Multi-Instance GPU and Multi-Process Service for Deep Learning Clusters," in *Proceedings of ICPP 2024*, 2024, pp. 504–513. [Online]. Available: https://doi.org/10.1145/3673038.3673089

[2] Z. Xia, Y. Hao, J. Duan, C. Wang, and J. Jiang, "Towards Optimal Preemptive GPU Time-Sharing for Edge Model Serving," in *Proceedings of the 9th International Workshop on Container Technologies and Container Clouds (WoC 2023)*, 2024, pp. 13–18, woC 2023 event; export publication year 2024; printed reference year 2023. [Online]. Available: https://doi.org/10.1145/3631311.3632401

[3] J. Gu, Y. Zhu, P. Wang, M. Chadha, and M. Gerndt, "FaST-GShare: Enabling Efficient Spatio-Temporal GPU Sharing in Serverless Computing for Deep Learning Inference," in *Proceedings of ICPP 2023*, 2023, pp. 635–644. [Online]. Available: https://doi.org/10.1145/3605573.3605638

[4] D. Luo, T. Yu, Y. Wu, H. Wu, T. Wang, and W. Zhang, "SPLIT: QoS-Aware DNN Inference on Shared GPU via Evenly-Sized Model Splitting," in *Proceedings of ICPP 2023*, 2023, pp. 605–614. [Online]. Available: https://doi.org/10.1145/3605573.3605627

[5] J. Chen, Y. Liang, B. Gao, Z. Zhou, F. Xu, X. Chen, and D. Guo, "MIGServe: Layout-Aware Multi-Instance GPU Management for Efficient LLM Serving," in *Proceedings of ICPP 2026*, 2026, pp. 844–854. [Online]. Available: https://doi.org/10.1145/3832810.3832898

[6] Z. Jiang, Z. Cai, Y. Li, Z. Wang, R. Ma, H. Guan, and R. Buyya, "MMK: A Hybrid Scheduling Framework for Fine-Grained GPU Sharing for Deep Learning Applications," *ACM Transactions on Architecture and Code Optimization*, vol. 23, no. 3, pp. 1–25, 2026, art. no. 84. [Online]. Available: https://doi.org/10.1145/3820163

[7] B. Zhang, Y. Wang, and Z. Li, "BOER: Enhancing Resource Utilization for Deep Learning Inference with Hybrid Spatial GPU Sharing," in *Proceedings of SC 2025*, 2025, pp. 519–532. [Online]. Available: https://doi.org/10.1145/3712285.3759857

[8] S. Li, T. Hanawa, and Y. Miki, "Adaptive GPU Sharing for Real-time LLM Serving with Best-effort Workloads," in *ICPP Workshops 2026*, 2026, pp. 1–6. [Online]. Available: https://doi.org/10.1145/3816891.3834895

[9] Y. Zhang, H. Yu, C. Han, C. Wang, B. Lu, Y. Li, Z. Jiang, Y. Li, X. Chu, and H. Li, "SGDRC: Software-Defined Dynamic Resource Control for Concurrent DNN Inference on NVIDIA GPUs," in *Proceedings of PPoPP 2025*, 2025, pp. 267–281. [Online]. Available: https://doi.org/10.1145/3710848.3710863

[10] P. Sinha, D. Liakopoulos, N. Lemma, and N. J. Yadwadkar, "Beyond Utilization: Energy-Conscious GPU Sharing for Inference Serving," in *Proceedings of SOSP 2026*, 2026, pp. 65–89. [Online]. Available: https://doi.org/10.1145/3830418.3843913

[11] X. Hui, Y. Xu, and X. Shen, "FluidFaaS: A Dynamic Pipelined Solution for Serverless Computing with Strong Isolation-based GPU Sharing," in *Proceedings of HPDC 2025*, 2025. [Online]. Available: https://doi.org/10.1145/3731545.3731580

[12] S. Zhang, Q. Chen, W. Cui, H. Zhao, C. Xue, Z. Zheng, W. Lin, and M. Guo, "Improving GPU Sharing Performance through Adaptive Bubbleless Spatial-

Temporal Sharing," in *Proceedings of EuroSys 2025*, 2025. [Online]. Available: https://doi.org/10.1145/3689031.3696070

[13] P. Elvinger, F. Strati, N. E. Jerger, and A. Klimovic, "Understanding GPU Resource Interference One Level Deeper," in *Proceedings of SoCC 2025*, 2025, printed SoCC 2025 event; export year 2026. [Online]. Available: https://doi.org/10.1145/3772052.3772270

[14] J. Hao, P. Subedi, L. Shramaswamy, and I. K. Kim, "Reaching for the sky: Maximizing deep learning inference throughput on edge devices with ai multi-tenancy," *ACM Transactions on Internet Technology*, 2023. [Online]. Available: https://doi.org/10.1145/3546192

[15] Z. Zhao, N. Ling, N. Guan, and G. Xing, "Miriam: Exploiting elastic kernels for real-time multi-dnn inference on edge gpu," in *Proceedings of ACM/IEEE Symposium on Edge Computing 2023*, 2023. [Online]. Available: https://doi.org/10.1145/3625687.3625789

[16] L. Ma, H. Chen, E. Shao, L. Wang, Q. Chen, and G. Tan, "Elasticroom: Multi-tenant dnn inference engine via co-design with resource-constrained compilation and strong priority scheduling," in *Proceedings of HPDC 2024*, 2024. [Online]. Available: https://doi.org/10.1145/3625549.3658654

[17] X. Wu, J. Rao, W. Chen, and H. Huang, "Switchflow: Preemptive multitasking for deep learning," in *Proceedings of Middleware 2021*, 2021. [Online]. Available: https://doi.org/10.1145/3464298.3493391

[18] Y. Zhang, Z. Zhang, W. Bao, and D. Yuan, "Itif: Integrated transformers inference framework for multiple tenants on gpu," in *Proceedings of ICPP 2023*, 2023. [Online]. Available: https://doi.org/10.1145/3605573.3605585

[19] V. M. Bhasi, A. Sharma, R. Jain, J. R. Gunasekaran, A. Pattnaik, M. T. Kandemir, and C. Das, "Towards slo-compliant and cost-effective serverless computing on emerging gpu architectures," in *Proceedings of Middleware 2024*, 2024. [Online]. Available: https://doi.org/10.1145/3652892.3700760

[20] A. Jahanshahi, M. Rezvani, and D. Wong, "Wattwiser: Power & resource-efficient scheduling for multi-model multi-gpu inference servers," in *Proceedings of IGSC 2023*, 2023. [Online]. Available: https://doi.org/10.1145/3634769.3634807

[21] Y. Chen, W. Li, H. Zhou, X. Yang, and Y. Yin, "Deinfer: A gpu resource allocation algorithm with spatial sharing for near-deterministic inferring tasks," in *Proceedings of ICPP 2024*, 2024. [Online]. Available: https://doi.org/10.1145/3673038.3673091

[22] X. Hui, Y. Xu, Z. Guo, and X. Shen, "Esg: Pipeline-conscious efficient scheduling of dnn workflows on serverless platforms with shareable gpus," in *Proceedings of HPDC 2024*, 2024. [Online]. Available: https://doi.org/10.1145/3625549.3658657

[23] M. Han, R. Chen, W. Shen, H. Zhang, J. Yang, and H. Chen, "Real-time, Work-conserving GPU Scheduling for Concurrent DNN Inference," *ACM Transactions on Computer Systems*, vol. 44, no. 1, pp. 1–42, 2025, art. no. 1; publication date November 2025 as printed in supplied report. [Online]. Available: https://doi.org/10.1145/3768622

[24] S. Choi, S. Lee, Y. Kim, J. Park, Y. Kwon, and J. Huh, "Serving Heterogeneous Machine Learning Models on Multi-GPU Servers with Spatio-Temporal Sharing," in *USENIX*

ATC, 2022, pp. 199–216. [Online]. Available: https://www.usenix.org/conference/atc22/presentation/choi-seungbeom

[25] Z. Bai, Z. Zhang, Y. Zhu, and X. Jin, "PipeSwitch: Fast Pipelined Context Switching for Deep Learning Applications," in *USENIX OSDI*, 2020, pp. 499–514. [Online]. Available: https://www.usenix.org/conference/osdi20/presentation/bai

[26] A. Gujarati, R. Karimi, S. Alzayat, W. Hao, A. Kaufmann, Y. Vigfusson, and J. Mace, "Serving DNNs like Clockwork: Performance Predictability from the Bottom Up," in *USENIX OSDI*, 2020, pp. 443–462. [Online]. Available: https://www.usenix.org/conference/osdi20/presentation/gujarati

[27] M. Yu, A. Wang, D. Chen, H. Yu, X. Luo, Z. Li, W. Wang, R. Chen, D. Nie, H. Yang, and Y. Ding, "Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference," in *USENIX ATC*, 2025, pp. 597–612. [Online]. Available: https://www.usenix.org/conference/atc25/presentation/yu

[28] H. Wu, Y. Yu, J. Deng, S. Ibrahim, S. Wu, H. Fan, Z. Cheng, and H. Jin, "StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow," in *USENIX ATC*, 2024, pp. 59–73. [Online]. Available: https://www.usenix.org/conference/atc24/presentation/wu-hao

[29] S. Zhang, A. Xu, Q. Chen, H. Zhao, W. Cui, Z. Wang, Y. Li, L. Xiao, and M. Guo, "KRYPTON: Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception," in *USENIX ATC*, 2025, pp. 1003–1019. [Online]. Available: https://www.usenix.org/conference/atc25/presentation/zhang-shulai

[30] J. Wang, Y. Wang, M. Han, and R. Chen, "SIRIUS: Colocating ML Inference and Training with Fast GPU Memory Handover," in *USENIX ATC*, 2025, pp. 1657–1675. [Online]. Available: https://www.usenix.org/conference/atc25/presentation/wang-jiali

[31] W. Zhao, A. Jayarajan, and G. Pekhimenko, "Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads," in *ACM ASPLOS Volume 1*, 2025, pp. 1052–1068. [Online]. Available: https://doi.org/10.1145/3669940.3707282

[32] M. Lee, S. Seong, M. Kang, J. Lee, G.-J. Na, I.-G. Chun, D. S. Nikolopoulos, and C.-H. Hong, "ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments," in *IEEE/ACM SC*, 2024. [Online]. Available: https://ieeexplore.ieee.org/document/10793202

[33] F. Strati, X. Ma, and A. Klimovic, "Orion: Interference-aware Fine-grained GPU Sharing for ML Applications," in *ACM EuroSys*, 2024, pp. 1075–1092. [Online]. Available: https://doi.org/10.1145/3627703.3629578

[34] A. Dhakal, S. G. Kulkarni, and K. K. Ramakrishnan, "GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform," in *ACM SoCC*, 2020, pp. 492–506. [Online]. Available: https://doi.org/10.1145/3419111.3421284

[35] M. Chow, A. Jahanshahi, and D. Wong, "KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers," in *IEEE HPCA*, 2023, pp. 624–637. [Online]. Available: https://doi.org/10.1109/HPCA56546.2023.10071121

[36] W. Shen, M. Han, J. Liu, R. Chen, and H. Chen, "XSched: Preemptive Scheduling for Diverse XPUs," in *USENIX OSDI*, 2025, pp. 671–692. [Online]. Available: https://www.usenix.org/conference/osdi25/presentation/shen-weihang

[37] K. K. W. Ng, H. M. Demoulin, and V. Liu, "Paella: Low-latency Model Serving with Software-defined GPU Scheduling," in *ACM SOSP*, 2023, pp. 595–610. [Online]. Available: https://doi.org/10.1145/3600006.3613163

[38] C. Savard, N. Manganelli, B. Holzman, L. Gray, A. Perloff, K. Pedro, K. Stenson, and K. Ulmer, "Optimizing High-Throughput Inference on Graph Neural Networks at Shared Computing Facilities with the NVIDIA Triton Inference Server," *Computing and Software for Big Science*, vol. 8, 2024, art. no. 14. [Online]. Available: https://link.springer.com/article/10.1007/s41781-024-00123-2

[39] L.-X. Ma, L.-P. Wang, E. Shao, R.-Y. Cao, and G.-M. Tan, "VastPipe: A High-Throughput Inference System via Adaptive Space-Division Multiplexing for Diverse Accelerators," *Journal of Computer Science and Technology*, vol. 40, no. 2, pp. 444–463, 2025. [Online]. Available: https://doi.org/10.1007/s11390-024-3773-5

[40] J. Lou, Y. Sun, J. Zhang, H. Cao, Y. Zhang, and N. Sun, "ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs," *CCF Transactions on High Performance Computing*, vol. 5, pp. 304–321, 2023. [Online]. Available: https://doi.org/10.1007/s42514-023-00154-y

[41] T. Adufu and Y. Kim, "Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs," *Cluster Computing*, vol. 28, 2025, art. no. 608. [Online]. Available: https://doi.org/10.1007/s10586-025-05284-2

[42] F. Xu, J. Xu, J. Chen, L. Chen, R. Shang, Z. Zhou, and F. Liu, "iGniter: Interference-Aware GPU Resource Provisioning for Predictable DNN Inference in the Cloud," *IEEE Transactions on Parallel and Distributed Systems*, vol. 34, no. 3, pp. 812–827, 2023. [Online]. Available: https://doi.org/10.1109/TPDS.2022.3232715

[43] Y. Peng and H. Peng, "InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems," *Future Generation Computer Systems*, vol. 150, pp. 10–20, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167739X23003187